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Impact of Appliance Information on Short-Term Load Forecasting using Ensemble Models

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Impact of Appliance Information on Short-Term Load Forecasting using Ensemble Models

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Bachelor's Thesis

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Prof. dr. Kerstin Bunte (Intelligent Systems, University of Groningen)
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Prof. dr. Michael H.F. Wilkinson (Intelligent Systems, University of Groningen)

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Abstract

Short term load forecasting (STLF) has been widely researched and investigated in recent years, as it ties into the aim of creating more robust, efficient, and clean power grids. STLF is an essential part of the operations of an electricity provider. Non-intrusive Load Monitoring (NILM), introduced in 1992, is the problem of identifying the consumption, or more commonly the status of household appliances (On/Off) at any given time. Many approaches have been undertaken and researched in an attempt to improve the accuracy of STLF, since even slight improvements could mean massive gains in terms of wasted energy, greenhouse gas emissions and, most importantly, operating expenses. This study investigates how appliance level information affects short term residential load forecasting when using three tree based regression models (XGBoost, LightGBM and Random Forest). This work makes use of two distinct datasets, a publicly available one, REFIT, as well as one provided by a Dutch company, Powerchainger. Initially, we break down these sets into data driven clusters in order to assess and compare the effects for different behaviorally defined groups of users. Then we explore the impact of appliance-level information, both at a neighborhood scale, as well as at the individual level. Finally, we perform an ablation study on the appliances so as to quantify and assess the contributions of each. Results indicate that moving from a single neighborhood model to aggregated household models with device features provides marginal and inconsistent gains. However, these findings vary a great deal based on the cluster, and therefore the behavior exhibited by the households that make it up, where for some user types the accuracy is clearly improved. Overall, including appliance level features mostly reduces the tendency towards over prediction, and is most beneficial when working with behaviorally complex and high-demand homes. Although these findings are clearly visible and observable when working with larger datasets, the trend remains visible with more limited in scope dataset with added appliance features. The obtained outcomes highlight that device contributions are highly dependent on data regularity and behavioral patterns, and varies meaningfully on a case by case basis. This study suggests that incorporating appliance information at scale in a targeted manner can provide significant improvements to short term load forecasting, paving the way for a less energy wasteful society and a more efficient grid.

Symbols

η	Learning rate
y_t	Original time series value at time index t
x_t	Differenced series at time t
∇	Differencing operator
∂	Partial derivative operator
d	Differencing order; number of times that differencing is applied
μ	Constant term
p	AR order; number of lagged values of x_t used
ϕ_i	Autoregressive coefficient for lag i
q	MA order; number of lagged error terms used
θ_j	Moving-average coefficient for lag j
ϵ_t	Random error at time t
T_b	Decision tree number b
m	Current iteration
M	Number of trees
t	Current time
$\hat{y}_t^{(m)}$	Prediction at iteration m for time t
X_t	Feature vector at timestep t
$\mathcal{J}$	Regularized objective function
$g_t^{(m)}$	First-order gradient of the loss with respect to the prediction
ℓ	Regression loss function; squared error
Ω	Regularization term that penalizes model complexity
D	Target day to be forecasted
A	Actual value
P	Predicted value
n	Size of validation sample
F	Number of sequential cross validation folds

Chapter 1

Introduction and Related Work

In a world where the negative impacts of climate change do not show any signs of slowing down and emission targets seem almost impossible to achieve; attaining a higher level of sustainability and energy efficiency in power utility networks is paramount [1]. One of the key components of attaining this goal is accurate *short-term load forecasting*(STLF). STLF is at the core of the operations of an energy provider and is essential for operational planning. Overestimating the demand leads to overproduction, waste and unnecessary costs, while underestimating the demand could lead to shortages, blackouts, and potentially severe financial and environmental consequences for communities and the economy [2, 3]. The basic principle here is to ensure a sufficient supply of energy to customers, which is essential to the daily operations of a power utility company. In this study, we will be focusing on STLF providing hourly predictions for the day-ahead (24h) with a lead time. Utility companies often need a forecasting lead time of at least half a day in order to prepare and adjust for the predicted demand, therefore, we will be using historical data up to the previous day at 10 o'clock in the morning to make predictions.

When it comes to the power grid, there are various groups of stakeholders; among whom, residential customers represent a significant portion of electricity consumption. According to the Central bureau for statistics of the Netherlands, they represent around 20% of total consumption [4]. STLF at this level is extremely difficult due to the high uncertainty in load demand, which is a non-linear and non-stationary time series. This demand is influenced by a variety of variables, such as weather conditions, calendar events, social activities, time of day, day of the week, etc. In addition, the rapid spread of distributed energy resources, such as renewables, or the proliferation of electrified transport have only added to the growing complexity of power systems.

In order to address these issues and help increase the accuracy of STLF and forecasting in general, engineers at utilities have been looking into Load Monitoring(LM) [5, 6], which focuses on measuring the set of appliances that are operating on at a customer's premises at any given point. However, this solution is not feasible when it comes to the residential setting, as it involves installing a sensor on each appliance. In contrast, Non-Intrusive Load Monitoring(NILM), operates by simply measuring the total power consumption at the household level, through which it then infers the load of individual appliances. Thanks to the lower complexity and costs of implementation, NILM methods are becoming very popular when it comes to building a customer's profile.

This paper proposes investigating how and if appliance-level features, which in the residential setting can be more readily made available through NILM, can improve electricity demand forecasting accuracy. The established practice for forecasting has relied primarily on aggregate load, as well as environmental and calendar variables [7, 8]. However, this study intends to also integrate appliance features in order to test and build appliance-aware forecasting models that focus on day-ahead predictions. In addition, this work is conducted in conjunction with a Dutch company in the field, namely, **Powerchainger**[1]. This study will be making use of the real-world customer data they have gathered.

There is no optimal model for solving any of the various load forecasting problems. Power utilities generally rely on proprietary and tailor made models for their specific local needs [9, 10]. A large variety of statistical models have been implemented in the past, with one such model, namely Autoregressive Integrated Moving Average Model (ARIMA) [11], which is still broadly used today. Classical statistical models, such as ARIMA, are popular and can capture cyclical patterns of energy consumption based on seasonal factors [12]. They have become the standard for a baseline comparison in most STLF papers examining various more novel approaches. This is due their extensive usage over the years as well as the well documented limitations of ARIMA and such models, namely their linear structure and therefore inability to account for exogenous variables with complex nonlinear dependencies [13].

Machine learning algorithms, by contrast, are able to handle both numerical features, such as temperature for instance, as well as categorical ones like day of the week, or holidays, etc. However, if not tuned properly these algorithms are prone to overfitting.

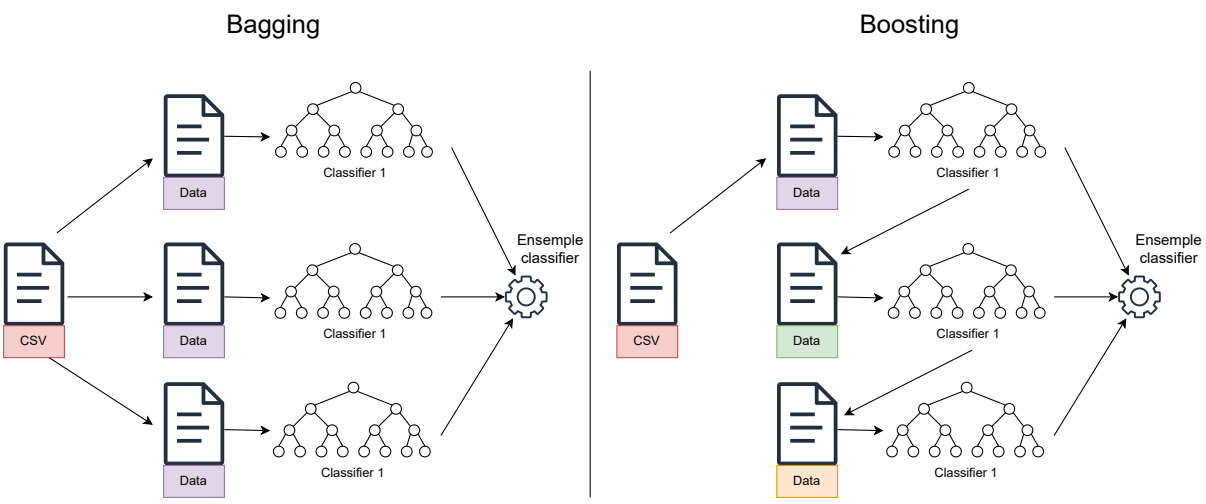


Figure 1.1: Bagging and Boosting

Ensemble models combine multiple base model predictions in order to improve accuracy. Some common and popular ensemble methods rely on bagging and boosting of decision trees. Boosting is an ensemble technique that builds the trees sequentially with residuals of the previous decision trees as the learning target. These processes are illustrated in the figure 1.1. These traditional machine learning approaches remain quite popular as they are simple, computation-

[1] https://powerchainger.io/

ally efficient and highly interpretable, while still maintaining a competitive performance in the domain [14, 15].

Over the past few years, one specific gradient boosting framework has been attracting a lot of attention when it comes to STLF, that being Extreme Gradient Boosting (XGBoost) [16]. It has been broadly used in a myriad of various fields and applications [17, 18, 19, 20]. XGBoost incrementally builds additive regression trees and uses regularization in order to minimize the loss function and control overfitting [14, 21]. It has been shown that if the data processing is rigorous, and the feature selection and engineering are properly addressed, then XGBoost can perform at a very high standard [21].

Another gradient-boosting framework that has recently shown promise in STLF is Light Gradient Boosting Machine (LGBM), introduced by Microsoft in 2016 [22]. In order to reduce the training complexity and improve convergence, this framework relies on gradient-based one side sampling (GOSS) and exclusive feature bundling (EFB). EFB is used to reduce the number of features by merging the exclusive ones, while GOSS retains all the data with a large gradient and a portion of low gradient data. The combination of these helps reduce the search space. When it comes to load forecasting, LightGBM has shown a lot of promise, being used either as a standalone model or as part of a hybrid model. In these studies and experiments, LightGBM has been found to outperform many other standalone models, including neural networks based candidates [23].

In the bagging approach to ensemble models based on decision trees, multiple training subsets are created by randomly sampling the original data with replacement. Random Forest (RF), is a tree-based bagging model, which has been widely used for STLF [24]. RF builds many decision trees on bootstrap samples, chooses a random subset of features at each split and averages the resulting predictions in order to control and mitigate overfitting. As opposed to XGBoost, this approach builds the tress in a parallel manner. Random forest has been shown to outperform many other models, for instance regression trees and SVM, when it comes to hourly electricity consumption in buildings [25]. When focusing on predicting the consumption in electric vehicles, and working with machine learning methods, RF and XGBoost have outperformed all other in the study [26].

These bagged and boosted tree-based approaches have shown good performance in many different studies regarding STLF, and are usually implemented as individual models. However, some studies have investigated their performance as part of a hybrid model or stacked ensemble. These papers have shown that by working with properly tuned models and combining their predictions, the performance metrics can be further improved [10, 27].

Non-intrusive load monitoring (NILM) has grown in popularity and interest in recent years, particularly as power utilities try to find cost-effective techniques to reduce waste and improve their overall efficiency. This is due to its relatively low cost of implementation paired with its large potential impact, owing to its ability to provide a much more detailed user profile and add extra features to consider. The impact of such features and insights shows promise, however, it has yet to be researched and tested further especially outside of the very short term load forecasting [28, 29].

Our main contributions are, as follows:

O1) The analysis of the impact of appliance features on Ensemble models

O2) The comparison of a few appliance-aware classical machine learning models

This paper is structured as follows. Initially, discuss data collection and manipulation in chapter 2. We outline the methodology utilized for our proposed experiments and analysis in chapter 3. Then we present all of our experiments, accompanied by the results and a discussion in chapter 4. Finally, in chapter 5 we give a general overview of the findings of this paper and explore the limitations of this study and debate potential future work.

Chapter 2

Data

2.1 Data Collection & Pre-processing

In this section we present in great details the data that we use and its particularities. Firstly, we define the collection process for each individual dataset in section 2.1.1. Then, we illustrate the pre-processing steps necessary to manipulate the data and present it in an appropriate format for the experiments in section 2.2. Subsequently, we describe the used methodology for analyzing the load consumption regularity exhibited by each household in order to group them into behavior-driven clusters in section 2.3. Lastly, we present the feature engineering steps undertaken and the final set of features in section 2.4.

2.1.1 Collection

In this study, we will be working with two sources of data. The datasets vary greatly in elapsed recorded duration, which provides an opportunity to study the effects that access to data has on ulterior prediction accuracy.

Table 2.1: Overview of employed datasets and their characteristics

Household	**Timeframe**	**Sampling**	**Location**
REFIT dataset (University of Glasgow)			
house_1	09-10-2013 – 10-07-2015	6–8 s	Loughborough
house_2	17-09-2013 – 28-05-2015	6–8 s	Loughborough
house_3	25-09-2013 – 02-06-2015	6–8 s	Loughborough
Powerchainger dataset			
house_1	13-12-2024 – 09-01-2025	1 s	Groningen
house_2	04-12-2024 – 03-01-2025	1 s	Groningen
house_5	29-04-2025 – 25-06-2025	1 s	Groningen

Table 2.1, presents a sample of the data at hand as well as some characteristics and information for each, we will dive into more details for both of the sources in each of the subsequent subsections 2.1.1 and 2.1.1.

REFIT

The first dataset to be used represents an open and publicly available dataset, built and collected via the project entitled Personalised Retrofit Decision Support Tools for UK Homes using Smart Home Technology (REFIT)[30]. The project was a collaboration among the Universities of Strathclyde, Loughborough and East Anglia [1]. We made use of a cleaned version made available by the University of Strathclyde in Glasgow [31].

The REFIT information provided contains consumption data for 20 smart homes in the Loughborough area, over the period ranging from 2013 to 2015. The information stored in the files consists of the total consumption `Aggregate` in Watts per hour, a timezone-aware timestamp `Time`, and the broken down consumption of multiple appliances, for a total of 9 appliances per household: `Appliance1`, `Appliance2`, ..., `Appliance9`.

Powerchainger

We also worked with a few proprietary residential electricity consumption datasets provided by Powerchainger. The datasets contain timezone-aware timestamps `time`, total consumption `main` and columns for device-specific consumption, such as: `kettle`, `oven`, `refrigerator`. Each dataset concerns one household, however the time period covered varies per house, in general the are around one month long with a couple covering two months. A subset of the best datasets has been chosen for the study, based on the amount of missing entries as well as the total number of data points.

2.2 Pre-processing

The series is stored into a pandas DataFrame indexed by Date, which is then aggregated to hourly resolution, resulting in an hourly series with columns `time/Time` and `Aggregate/main` (representing the total consumption), this served as the primary basis for model training and evaluation. This process is the same for both our sources of data.
Hourly meteorological information was compiled with the help of an open and publicly available API : Open Meteo[2], which provides historical data given a time range and location coordinates. This method was also used to collect historical forecasts. The coordinates used to collect these data were the location of the Loughborough area for the REFIT dataset, as for those provided by Powerchainger, the location of Groningen (where the company is based) was used. The temperature information is merged with the load data on timestamps, so that each hourly load observation is matched with the corresponding ambient temperature. Additionally, in order to fill in missing values for the total consumption, we measured the mean usage at every hour per weekday, per month, and replace the unavailable values with these means.

[1] https://pureportal.strath.ac.uk/en/datasets/refit-electrical-load-measurements-cleaned/
[2] https://open-meteo.com/en/docs

2.3 Data Analysis

Residential load consumption depends primarily on user behavior and therefore varies greatly from one household to the next. This makes working with individual dwelling's electricity usage quite difficult in comparison to a substation level, since consistent and obvious patterns are hard to come by. The aggregate load of many households allows for the smoothing over of the daily usage or profile, while the consumption of an individual residential customer is heavily influenced by the underlying human behavior and number of residents for each house. As such the daily profiles and patterns of each can vary, certain households have more consistent consumption compared to others who are more impulsive. Therefore, for each resident, their lifestyle as well as the type of appliances they use will have a major impact on the load profile of each.

This phenomenon is well documented and known to power utilities who in turn build various profiles of different types of users in an effort to better estimate the actual consumption of different households, this can be observed when a new customers signs a contract with a electricity provider. The utility company usually presents an estimate of the monthly cost for the customer based on various factors such as their age, occupation, the size of the household, etc. However, the diversity and complexity of human behavior yields a massive variety in the load of different houses.

Consequently, we have grouped different houses into clusters of varying difficulty in terms of predictability based on their behavior and usage in general. This grouping was achieved through the use of a density-based clustering approach, namely, Density Based Spatial Clustering of Application with Noise (DBSCAN) [32]. This approach allows us to evaluate the consistency of the load of different households in our dataset, doing so without an explicit number of clusters. Which in turn also allows us to better understand our results and performance based on how consistent any household is.

DBSCAN works with three predefined parameters, namely, the minimum amount of samples in a cluster, the distance or neighborhood radius, and finally the distance metric used to calculate the distance between points. For our study we have made use of the sklearn.cluster library and set the maximum distance to 10% of the average daily load, the minimum amount of samples to 2 and lastly, the metric chosen is Euclidean distance. This DBSCAN is done for each household and computes the number of outliers, the numbers of clusters the algorithm found for each house and the average daily usage [33]. Once this step is done, we then proceed to cluster the users together based on those metrics, namely the fraction of outliers, the number of clusters and the average daily consumption identified per household, for this run the maximum distance is determined using the K-distance heuristics[34]. In order to determine this neighborhood radius, we computed the distance to the K-th nearest neighbor in this three dimensional feature space, sorted these distances and analyzed the resulting plot, adopting its elbow as our maximum distance value. Finally, we obtain our groups and we then choose a representative household for each cluster, which we will then use in the experiments section to analyze the performance of our models.

2.4 Feature Engineering

All models were trained using the same engineered features. For each hourly timestamp, several calendar-derived features were constructed. These consist of the calendar year, month, day of the month, hour of the day and weekday (Monday coded as 0 and Sunday as 6). In order to capture systematic differences between weekdays and weekends, we defined an additional variable TypeDay, which takes the value 0 for workdays and 1 for weekends. The ambient temperature described in the collection process above was incorporated as a continuous covariate, aligned at the hourly level.

For the purpose of providing the models with a history of past load while respecting operational information constraints, we have made use of an explicit temporal offset to create lagged versions of the target variable at the day level. Specifically, once calendar features were added, the data was re-indexed and a set of lagged features lag-1 through lag-7 were generated using time-based shifts. For each row with the timestamp t, $lag - k$ has been defined as the load $Volume$ at $t - d_k$ days, where $d_k = 2 + (k - 1)$. Therefore, $lag - 1$ represents the load two days before, up to $lag - 7$ being eight days prior. This offset was selected in order to respect our setup requirements and the gap in the prediction process, that being the fact that when forecasting for a given day D, the models never use consumption values from after 10:00 on day $D - 1$. Consequently, the final feature set consists of calendar covariates:

- Year
- Day
- Hour
- Weekday
- TypeDay
- Temperature
- Lagged consumption features (lag-1 to lag-7)

In addition, when incorporating appliance features for every household, we convert the original signal into a binary On and Off one, which is represented using 0 and 1 respectively. In order, to perform this task while still avoiding appliance noise and focusing on distinct On and Off events, we use visual analysis to determine a threshold for each device above which the appliance is classified as On.

Chapter 3

Methods

In this chapter, we dive deeper into the theoretical foundation employed into conducting this project and confer an extensive overview depicting the methodology. Firstly, we discuss the main statistical model, ARIMA, used to create a performance baseline in section 3.1. Then, we provide a comprehensive analysis over the three main ensemble models: Random Forest, XBGBoost, and LightGBM in sections 3.2.1, 3.2.2 and 3.2.3 respectively. Finally, we examine the forecasting setup and the mechanism of hyperparameter optimization employed in sections 3.3.1 and 3.3.2 respectively.

3.1 Autoregressive integrated moving average (ARIMA)

Broadly used in the field to this day, for the purpose of time series forecasting, is Autoregressive Integrated Moving Average. The model assumes that the predicted value is dependent on a linear function of several previous values of the time series and some random errors. The first step undertaken by the model, is the calculation of the integrated component, this is achieved through subtracting each value from the previous retrospective value. The purpose of this process is to remove the trend and create a stationary time series. The common notation for this model is $ARIMA(p, d, q)$, where p is the lag order for the AutoRegressive model (representing the amount of prior periods of the time series are made use of), d indicates the order of differencing needed to achieve a stationary time series. Finally q refers to the lag count in the error term, often named the order of moving average. Let x_t denote the differenced time series at time t and y_t the original time series values at time t. Concretely, ARIMA first applies differencing d times: $x_t = \nabla^d y_t$, where $\nabla y_t = y_t - y_{t-1}$ in order to obtain an approximately stationary series. We can denote the ARIMA stationery series calculation as follows:

$$x_t = \mu + \sum_{i=1}^{p} \phi_i x_{t-i} + \epsilon_t + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} \tag{3.1}$$

In the previous equation, we use μ to represent a constant term used for the calculation, ϕ_i is the autoregressive coefficient for lag i, θ_j symbolizes the moving-average coefficient for lag j and, finally, ϵ_t illustrates a random error at time t, typically assumed with mean 0 and constant variance.

The main disadvantage of working with ARIMA is its inability in its standard form to take into consideration exogenous variables, as opposed to the quite flexible tree-based models for instance. This could be mitigated with the use of variants, such as SARIMAX. However, this is out of the scope of this study, where ARIMA only serves as a baseline.
The optimization of the main parameters is achieved through the use of a tuning helper function.

3.2 Ensemble models

The following subsections will discuss each of the ensemble models used in the study.

3.2.1 Random Forest (RF)

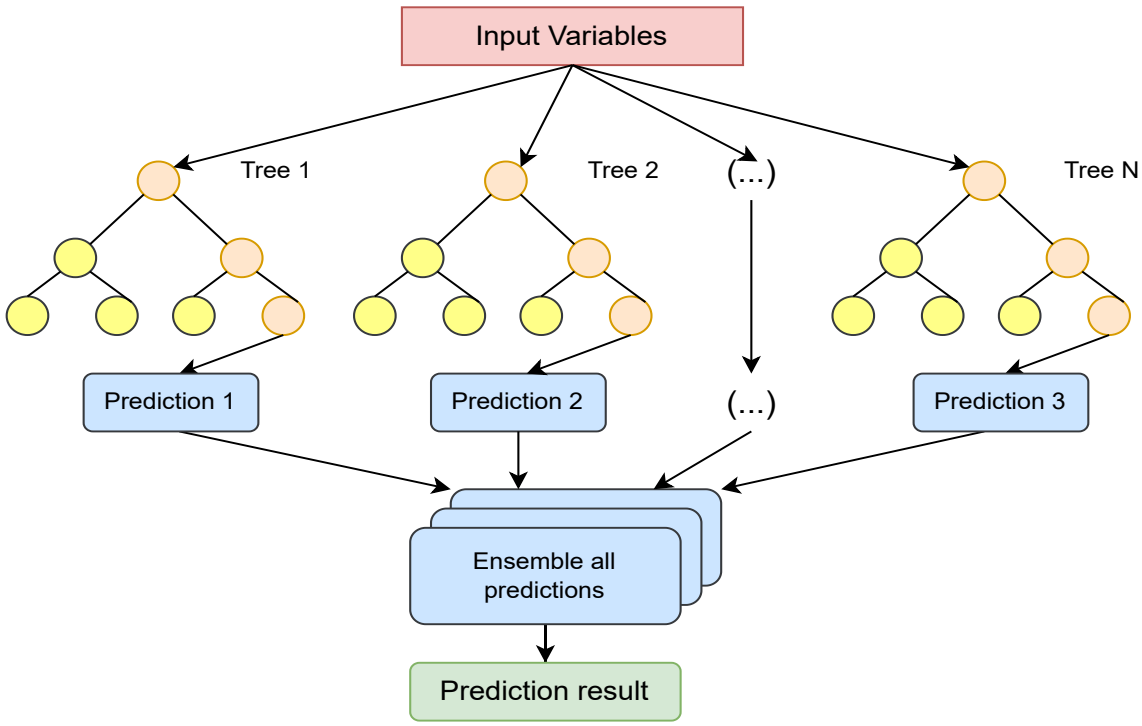


Figure 3.1: Simple Random Forest Overview

Random Forest has been a popular model in the machine learning field for a relatively long time. Due to its longevity, there exist a few different versions of it, such as Random Greedy Forest (RGF) that relies on regularization and employing a greedy structured learning approach instead of utilizing inherent randomness and variance reduction. However, for the purpose of this study, we will only be focusing on the original variant. Random Forest acts as a variation of bagging techniques with slight modifications. Unlike standard bagging, RF extracts a subset of features in order to train multiple trees, which makes the trees more independent. Random forest makes use of bootstrapping, to create a random subset of features that then

serves as the training set for an individual decision tree. This process harnesses the strength of multiple models as well as randomization to improve performance of the model while helping with avoiding overfitting. The general workflow for this algorithm is showcased in figure 3.1.

Random Forest operates by training M decision trees T_m, with $m = 1, 2, ..., M$, on bootstrap-resampled datasets and aggregates their predictions through the averaging function. For each iteration, the resulting function is applied on the feature vector at time step t. The corresponding formula can be denoted as:

$$\hat{y}_t = \frac{1}{M} \sum_{m=1}^{M} T_m(X_t) \tag{3.2}$$

The function we selected as the criterion for evaluating the quality of splits is the default, squared error. For all our ensemble models, we conducted hyperparameter optimization on the relevant parameters, with the help of our grid search function, as explained in the section 3.3.2.

3.2.2 Extreme Gradient Boosting (XGBoost)

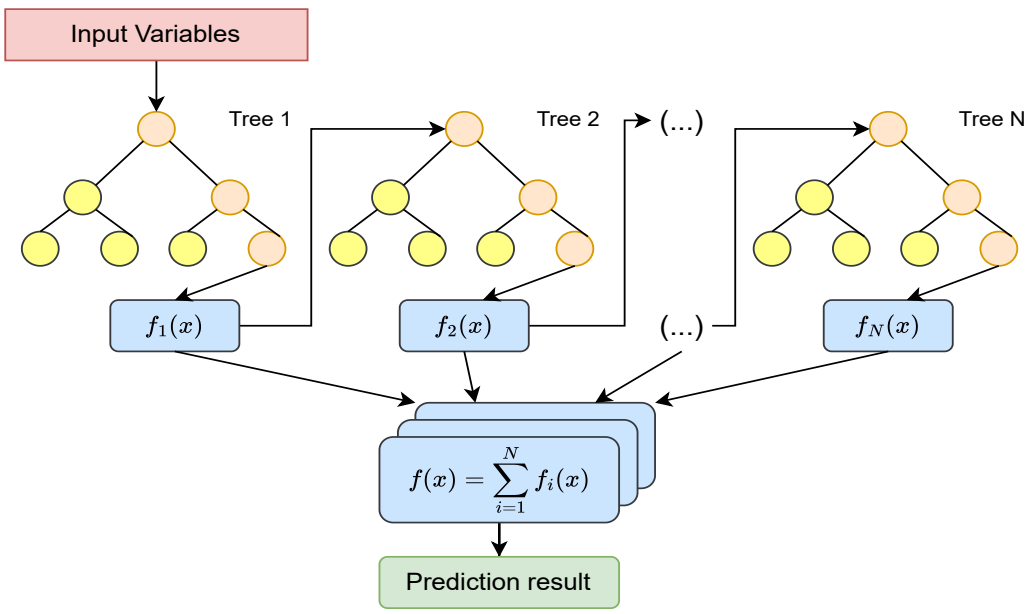


Figure 3.2: Simple XGBoost Overview

Introduced in 2016 by Tianqi Chen and Carlos Giestrin [16], XGBoost is a scalable gradient boosting framework that employs decision trees as its base model. The architecture leverages level-wise (depth-wise) growth, resulting in more balanced trees. The typical XGBoost workflow can be observed in figure 3.2. The framework uses loss functions to avoid overfitting and improve overall performance. Based on the original principle of boosting by Friedmann [35], it is a supervised learning approach. As compared to more traditional boosting techniques, XGBoost offers faster execution times as well as superior prediction accuracy. Let f_m refer to

the regression tree and η be a chosen learning rate. At each iteration step t, the model performs an update in the form of a learning step:

$$\hat{y}_t^{(m)} = \hat{y}_t^{(m-1)} + \eta\, f_m(X_t) \tag{3.3}$$

In order to make its final prediction, XGBoost generates a new tree that best corrects the current errors and combines itself with the use of a gradient descent method to reduce the loss. This is done by training the model to move predictions in the direction that reduces the loss, given by the negative gradient of the loss function with respect to the current prediction, shown in equation 3.6. The purpose of this approach is to minimize the regularized objective of the form depicted in figure 3.4. Since we use the squared error as the loss function, the gradient function is simplified to the version in equation 3.7.

$$\mathcal{J} = \sum_{t=1}^{T} \ell(y_t, \hat{y}_t) + \sum_{m=1}^{M} \Omega(f_m) \tag{3.4}$$

$$\ell(y_t, \hat{y}_t) = (y_t - \hat{y}_t)^2 \tag{3.5}$$

$$g_t^{(m)} = \left.\frac{\partial \ell(y_t, \hat{y})}{\partial \hat{y}}\right|_{\hat{y}=\hat{y}_t^{(m-1)}} \tag{3.6}$$

$$g_t^{(m)} = -2\left(y_t - \hat{y}_t^{(m-1)}\right) \tag{3.7}$$

We define ℓ as the regression loss, which we implement as the squared error function and can be visualized in equation 3.5. Furthermore, we denote Ω as a regularizing term that penalizes the complexity of the model to avoid overfitting. This process is then repeated, and the tree gets appended until no more additional enhancements can be made.

The objective function selected is the squared error regression, and training was performed on the CPU. Hyperparameter optimization was conducted for each model using a grid search function, this processed is explained in more details below in section 3.3.2, while the remaining parameters were left at library defaults.

3.2.3 Light Gradient Boosting Machine (LightGBM)

Published in 2016 by Microsoft Research Asia [22], the Light Gradient Boosting Machine makes use of decision trees just as XGBoost, it is also based on the boosting principle. However, it is characterized by leaf-wise growth, which can be observed in figure 3.3, that promotes faster training loss minimization. Due to the applied boosting strategy, it leverages an additive model that is the sum of M trees, where an update step takes the form of:

$$\hat{y}_t^{(m)} = F_m(X_t) = F_{m-1}(X_m) + \eta f_m(X_t), \qquad \text{with } F_0(X_t) = const \tag{3.8}$$

Similarly to XGBoost, the model leans stage-wise by minimizing a regularized objective, such as in equation 3.4. However, it introduces two new techniques, namely, gradient one-sided sampling technique and independent feature merging technique. Some of LightGBM's main advantage is high training speed, while making use of minimal memory, this makes it very efficient and able to process batch data quickly.

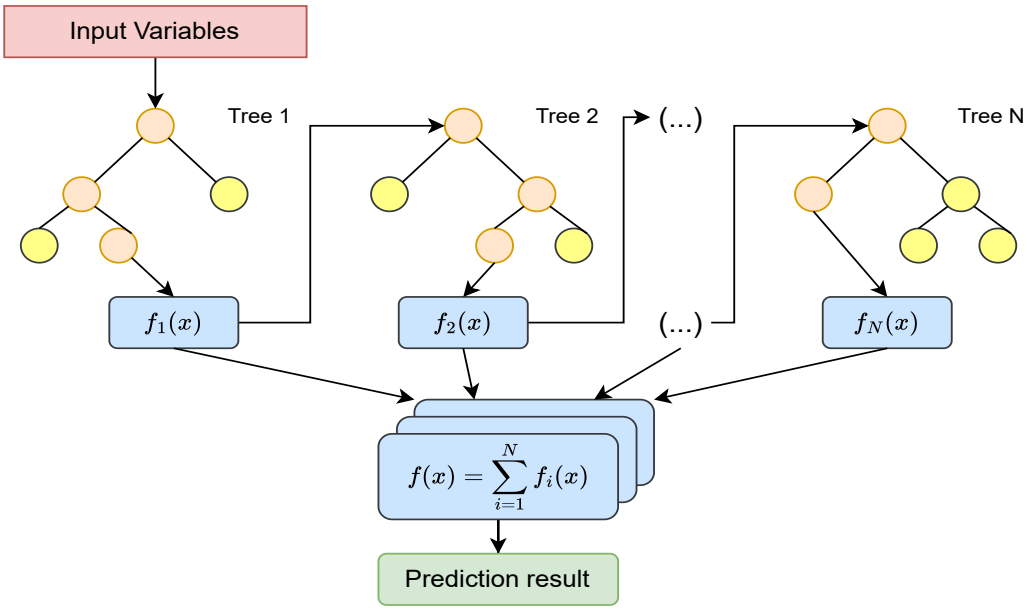


Figure 3.3: Simple LightGBM Overview

Key hyperparameters were optimized using a grid search function, just as the XGBoost model. The model was trained on the CPU with multi-threading enabled.

3.3 Setup

3.3.1 Forecasting setup

The prediction task is formulated as a day-ahead forecasting problem with an hourly resolution and a 14-hour lead time. Therefore, for a target day D, the models are trained only on information up to the prior day $D-1$. Then we make the hourly prediction for the next two weeks, where for every day D, the model receives as input data up to $D-1$ at 10 am. The forecasting procedure is the same for all the ensemble models (XGB, RF and LGBM), as well as the baseline model (ARIMA), facilitating direct comparison of the performance of our models under the same information constraints and feature set.
A helper routine constructs a 24 hour sequence of timestamps for day D, which is then merged with historical hourly forecast values. Calendar-derived features (Year, Month, etc) are subsequently computed for the 24 timestamps of day D. For the purpose of generating our lagged features while respecting the information constraints, we make use of our cutoff (10:00 on day $D-1$) to limit the historical data concatenated with the 24 hour forecast template into a single time series. The resulting matrix is then passed along to each model so as to obtain the load prediction for the day D. These predictions are subsequently merged with the actual observed hourly load for the same timestamps with the goal of forming an evaluation dataset.
In the field of load forecasting the main metrics that are commonly considered to measure the performance of a model or the quality of its predictions are the mean absolute error and the

mean absolute percentage error, which we shall also be employing in this study. Where Ai and Pi are the actual and predicted values, respectively; and n is the size of the validation sample :

- Mean Absolute Error (MAE) :

  $\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|A_i - P_i|$

- Mean Absolute Percentage Error (MAPE) :

  $\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\frac{|A_i - P_i|}{A_i}$

3.3.2 Hyperparameter Optimization

Hyperparameter optimization was performed on a time-series aware cross-validation structure, implemented with sckit-learn's TimeSeriesSplit. The available historical data up to the cutoff were split into F sequential folds, where each fold is made up of a training window as well as a validation window. This process allows the model to train on earlier observations and be evaluated on a later block of data, which in turns preserves the temporal order and prevents the leakage of future data. The number of splits is determined depending on the length of the available training period.
For each model, a grid of hyperparameter values is defined and investigated using sckit-learn's GridSearchCV, with TimeSeriesSplit as the cross-validation technique. Where all specified combinations of hyperparameters are evaluated. For each parameter configuration, the model is fitted and its predictive performance is computed according to a scoring metric, which we define as the mean absolute error. For every single model, the configuration that achieved the best average performance across the time series splits is selected. The corresponding model was refit on the full training set, up to the cutoff, before then being used to perform the final 24-hour predictions for a target day. For each of the models we employed, specific hyperparemeters were tuned according to the model in question. The following tables 3.1, 3.2 and 3.3 show the parameters that were tested as well as the search space for each in combination with a small description.

Table 3.1: Search space for XGBoost hyperparameters

Hyperparameter	**Search Space**	**Description**
`# estimators`	$\{200\}$	Number of boosting rounds
`max depth`	$\{4, 6, 8, 10\}$	Maximum depth of each tree
`learning rate`	$\{0.05, 0.1\}$	Step size shrinkage
`subsample`	$\{0.8, 1.0\}$	Fraction of samples used per tree
`colsample by tree`	$\{0.8, 1.0\}$	Fraction of features used per tree

One of the limitations of this study was the limited computational time and available hardware resources, which required limiting the hyperparameter search and restricting the range and granularity of model configurations investigated during tuning.

Table 3.2: Search space for LightGBM hyperparameters

Hyperparameter	**Search Space**	**Description**
`# estimators`	$\{200\}$	Number of boosting rounds
`max depth`	$\{4, 5, 6\}$	Maximum depth of each tree
`learning rate`	$\{0.05, 0.01\}$	Step size shrinkage
`subsample`	$\{0.6, 0.7\}$	Fraction of samples used per tree
`colsample by tree`	$\{0.6, 0.7, 0.8\}$	Fraction of features used per tree
`# leaves`	$\{31\}$	Maximum number of leaves per tree
`minimum child samples`	$\{20, 50, 100\}$	Minimum samples required in a leaf

Table 3.3: Search space for Random Forest hyperparameters

Hyperparameter	**Search Space**	**Description**
`# stimators`	$\{200\}$	Number of trees
`max depth`	$\{5, 10, 15\}$	Maximum depth of each tree
`minimum samples for split`	$\{2, 5, 10\}$	Minimum samples required to split a node
`minimum samples at leaf`	$\{1, 2, 4\}$	Minimum samples required at a leaf node

Chapter 4

Experimental results and Discussion

In this section, we present our experiments with each of the data sources we have worked with. Initially, we quickly go over the software and hardware resources that were used to run all of the subsequent experiments in section 4.1. Furthermore, we discuss the clustering process of users based on the regularity of their usage, in section 4.2. Then, we explore the performance of our models when it comes to single house predictions versus aggregate predictions, simulating a neighborhood, in section 4.3. Subsequently, we analyze the experiment revolving around the effects of incorporating appliance information in section 4.4. Finally, we present an ablation study about the various appliance features available in each dataset, in section 4.5.

4.1 Software and Hardware

The following libraries and software were employed for this study : Python (v 3.10.12), Jupyter (v 7.4.7), Numpy (v 1.25.2), Pandas (v 2.3.3), Sckit-learn (v 1.7.2), XGBoost (v 3.1.1), LightGBM (v 4.6.0) and Statsmodels (v 0.14.5).
All of the calculations, training and forecasting were performed on the following hardware :

- Operating system: Windows Subsystem for Linux (WSL) v2
- CPU : Intel(R) Core(TM) i7-9750H
- GPU : NVIDIA GeForce RTX 2060
- RAM : DDR4, 16 GB

4.2 Clustering

In this initial experiment we focus on trying to group our households based on consistency of consumption and pick a representative household for each group as outlined earlier in section 2.3.

4.2.1 REFIT

Due to the incompleteness of the data we selected a year's worth of data points, with a small amount of missing values. These are then filled using the mean consumption for the specific month, weekday and hour of the day. The selected time period ranges from 2014-04-01 until

2015-04-01. As was expected, there is an extremely large variety in the amount of clusters and outliers DBSCAN finds for each house. A sample of these results are summarized in the following table 4.1. Additionally, DBSCAN will be used at the aggregated neighborhood level, and the results will be presented below.

Table 4.1: DBSCAN clustering statistics per REFIT household

Household	**Clusters**	**Days**	**Outliers**	**Outliers (%)**	**Avg. Daily Usage (Wh)**
House_1	4	365	106	29.04	10 311.07
House_2	14	365	189	51.78	11 063.13
House_4	4	365	25	6.85	8839.91
House_5	4	365	94	25.75	18 242.82
House_6	2	365	13	3.56	11 292.39
House_7	9	365	187	51.23	12 655.64
House_8	5	365	64	17.53	15 916.13
House_10	4	365	140	38.36	16 456.88
House_12	6	365	139	38.08	8965.21
House_13	6	365	115	31.51	13 222.98
House_15	6	365	53	14.52	6197.51
House_16	4	365	100	27.40	13 820.37
House_17	11	365	102	27.95	10 033.99
House_18	2	365	35	9.59	10 655.26
House_19	2	365	67	18.36	7029.39
House_21	5	365	76	20.82	13 492.18

The table shows a summary of the DBSCAN consistency analysis for each REFIT household. For every home, we constructed a daily matrix where every row corresponds to one day and each column an hourly reading. DBSCAN is then applied to the set of daily profiles. The number of clusters here refers to how many distinct clusters of "typical days" are detected for that household. Each cluster represents a recurrent pattern in daily electricity consumption, for instance a weekday and a weekend profile. The results show a wide spread in both the number of routine patterns identified as well as the degree of irregularity, represented by the number of outlier days.

The most consistent households, such as House_6 and House_4, only have two distinct clusters and a very small number fraction of outlier days, 3.6% and 6.9% respectively. This indicates that almost all their days can be described by one of two stable daily profiles. While on the other end of the spectrum, some of the most inconsistent houses are House_2 and House_7, where over half of their days are classified as outliers 51%. DBSCAN still identifies many distinct clusters for both homes however, the number is large, 14 and 9 respectively, suggesting that these daily patterns seldom repeat to create more cohesive clusters, making forecasting inherently more difficult. These results reinforce the idea that demand varies massively from one household to the next. In contrast, when working with an aggregate of all the houses combined we have much more consistent patterns and a very small amount of outlier days. The table below encapsulates the result of running DBSCAN on all houses combined, simulating the substation level 4.2.

The neighborhood was simulated by summing the hourly demand of all households and then

Table 4.2: DBSCAN clustering statistics for the simulated neighborhood

Clusters	Days	Outliers	Outliers (%)	Avg. Daily Usage (Wh)
1	365	1	0.27	390 219.84

constructing daily profiles in the same way as for the individual houses. In this case, DBSCAN only finds a single cluster of daily profiles and just one outlier day. This showcases how once individual household loads are aggregated, the resulting daily pattern is extremely consistent, almost every day is described by this dominant profile with only a single day deviating from it. In comparison to the high degree of heterogeneity observed at the individual level, the results presented in table 4.2 confirm the "smoothing" effect of aggregation, where idiosyncratic variations largely cancel out, yielding an electricity load that is much easier to forecast. Tabulated below 4.3 are the results of the second run of DBSCAN where the aim is to use the values found per household to form groups of similar users in terms of consistency. Therefore, each household was embedded into a three dimensional feature space defined by the fraction of outliers, the number of clusters and the average daily consumption. In order to determine the DBSCAN neighborhood radius we used the k-distance heuristic, so that the number of user groups was determined purely by the data. This differs from how power utilities would group users as we are not looking at daily behavior or age or any other factors that would definitely have an impact on the load.

Table 4.3: Summary statistics of resulting clusters

Cluster	Outliers (%)	Avg. # clusters	Avg. Daily Usage (Wh)
0	35.39	8.33	9866.82
1	30.86	5.13	15 150.24
2	10.77	2.83	9468.32

Table 4.3, presents a summary of the metrics pertaining to the clusters formed by DBSCAN. The three distinct clusters we are able to form can be summed up as:

- Cluster 0 : Irregular / atypical users
  Relatively normal usage, however a high number of outliers and large amount of day types.
  Houses : 1, 2, 7, 12, 15, 17

- Cluster 1 : High usage, multi routine users
  Slightly less outliers, higher consumption and still a medium amount of day types.
  Houses : 3, 5, 8, 9, 10, 13, 16, 21

- Cluster 2 : Regular users
  A much lower amount of outliers, "normal" consumption and a small amount of different day types.
  Houses : 4, 6, 11, 18, 19, 20

This information is visualized in figure 4.1, where the average daily usage is displayed in kWh and the outlier fraction is scaled up by a factor of 10, for better readability and easier comparison.

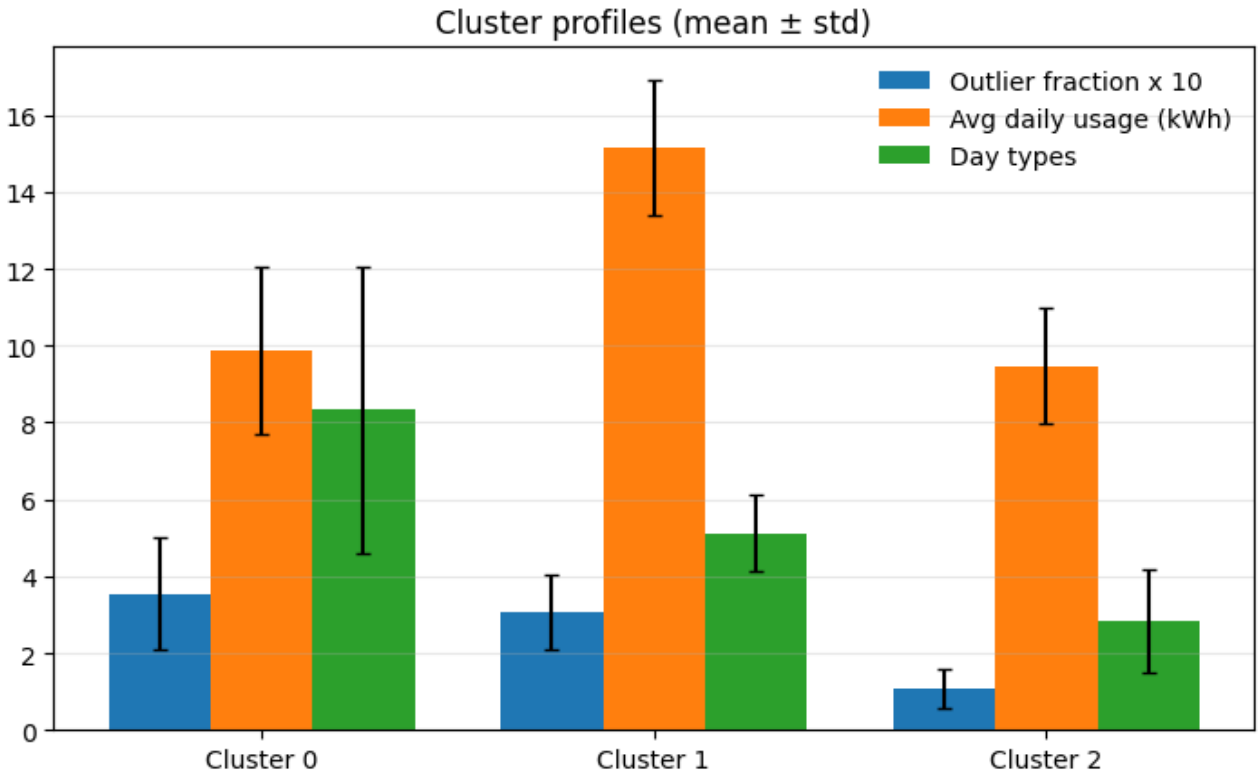


Figure 4.1: Visual representation of the User clusters obtained for the REFIT households

For each of these groups, we then select a representative house, which we will then use for the next experiments and model performance analysis. When looking at each cluster we use calculations such as taking the median when it comes to the fraction of outliers and pick a household that is closest to metrics of the cluster as a whole. The following table 4.4, contains the selected homes and their metrics.

Table 4.4: Selected representative households per cluster

Cluster	Household	Outliers (%)	# clusters	Avg. Daily Usage (Wh)
0	house_12	37.81	6	8939.57
1	house_13	31.51	6	13 222.98
2	house_18	9.59	2	10 655.26

The cluster 0 exemplar, house_12, has a relatively high outlier amount, about 38%, and six DBSCAN clusters, demonstrating the fragmented and irregular usage of this group. While the cluster 1 representative, house_13, also exhibits six routines but combines that with a slightly lower quantity of outliers (about 32%), and a higher average daily usage of about 13.2 kWh, making it a good fit for the high demand yet irregular group. Finally, the cluster 2 household,

house_18, has only two DBSCAN clusters and a very low amount of outliers at about 9.6%, reflecting a strong daily profile that repeat in a highly consistent way.

4.2.2 Powerchainger

Following the example set in the previous subsection 4.2.1, we will be analyzing the set of Powerchainger households and then attempting to fit them into coherent clusters based on user behavior. By contrast to the broad range and regular patterns of the REFIT dataset, the real-world data provided by the company showcases the inherent complex and random behaviors of individuals. The household datasets are collected over a much shorter time period, between one and three months, albeit not the same time interval and therefore spread out over a year. Due to this particularity of the data frames they can not be aggregated to simulate a neighborhood, because of the lack of a contiguous interval. We start by running DBSCAN on the chosen houses, in order to analyze the regularity of the behavior of each user.

Table 4.5: DBSCAN clustering statistics per Powerchainger household

Household	**Clusters**	**Days**	**Outliers**	**Outliers (%)**	**Avg. Daily Usage (Wh)**
House_0	5	63	22	34.92	11 103.57
House_1	1	28	17	60.71	3648.58
House_2	3	31	24	77.42	7489.11
House_3	1	30	25	83.33	15 326.38
House_4	1	27	13	48.15	12 207.97
House_5	2	58	49	84.48	6514.83
House_6	1	28	15	53.57	4286.17
House_7	6	80	39	48.75	2870.13
House_8	2	24	12	50.00	5230.59
House_9	1	37	8	21.62	7380.23

Table 4.5, reports the DBSCAN consistency metrics for the nine households provided by Powerchainger, which met the criteria of contiguous missing values being less than 5% of the total amount of the total amount of entries. The columns and metrics represent the same concepts as in the previous section 4.2.1. These households exhibit a very large amount of outliers, with DBSCAN identifying up to 83% of days as outliers in the worst case and 35% in the best case. In other words, for most of these users, the majority of days do not resemble any stable routine pattern, according to DBSCAN. For several houses, DBSCAN finds only a single cluster, despite a very large outlier fraction, suggesting that a small subset of days for a routine profile while the remainder behave independently. For others, a large number of clusters is identified, indicating a large mixture in daily usage patterns, however these are once again accompanied by a significant amount of outliers. Taken together, these households can be best characterized as behaviorally unstable with weak routine patterns, which will make forecasting more challenging.

In order to investigate the presence of distinct subgroups within this set, we apply a second stage clustering, in the same way as was done in section 4.2.1. The outcome of which is summarized in Table 4.6, where we can observe that all the houses fit into three groups based

Table 4.6: Summary statistics of resulting Powerchainger clusters

Cluster	Outliers (%)	Avg. # clusters	Avg. Daily Usage (Wh)
0	47.35	2.80	9777.66
1	54.76	1.33	4388.45
2	80.95	2.50	7001.97

on the selected metrics. In addition, figure 4.2 represents that same information in a visual manner, with the average daily usage converted into kWh and the outlier fraction scaled up by a factor of 10.

Cluster 0, exhibits a large amount of outliers, around 47%, as well as a moderate number of

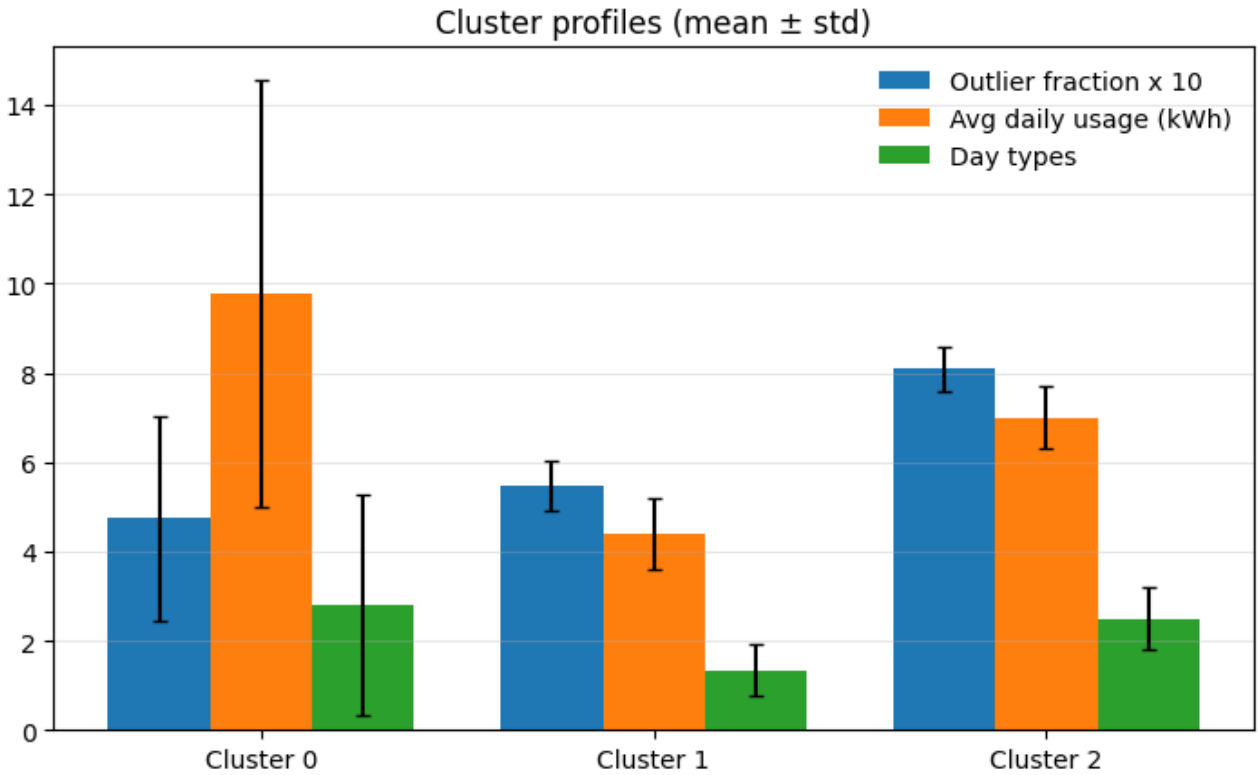


Figure 4.2: Visual representation of the User clusters obtained for the powerchainger households

day types (2.8) and a daily mean usage of about 9.7 kWh. This corresponds to the group with the least outliers out of all clusters, which contains some repeated structure that DBSCAN can capture. Cluster 1 on the other hand, has a comparable yet slightly higher outlier percentage of about 55%, while showing the lowest average daily usage of around 4.4 kWh as well as a lowest amount of day types of about 1.3. This suggest that this group of households displays relatively low demand and do not show many repeatable, stable daily routines. Finally, cluster 2, concentrates the least regular households as they have a mean outlier percentage of around 81%, as well as 2.5 day types and 7 kWh of daily consumption on average. This suggests that this group express a daily demand that is mostly non-repetitive over the observed period,

but where DBSCAN can still identify more than one recurring routine, indicating sporadic alternation between a small number of patterns in combination with many atypical days.

Table 4.7: Selected representative households per Powerchainger cluster

Cluster	Household	Outliers (%)	# clusters	Avg. Daily Usage (Wh)
0	house_9	21.62	3	7380.23
1	house_6	53.57	1	4286.17
2	house_2	77.42	3	7489.11

For the subsequent case study analysis, we select one representative households for each cluster, similarly as with the REFIT dataset, this selection is presented in table 4.7. House_9, with an outlier percentage of 22% and an average daily usage of about 7.4 kWh, is a relatively regular example within these households, represents cluster 0. For cluster 1, we have selected house_6 exhibiting a daily mean consumption of about 4.3 kWh, and about 54% outliers, highlighting the low usage and unstable nature of this group. Finally, we have chosen house_2 from cluster 2 for the subsequent experiments, with its high level of outliers (around 77%) and its moderate to high daily usage of around 7.5 kWh, to represent the most irregular households.

4.3 Aggregate vs Single Household Prediction

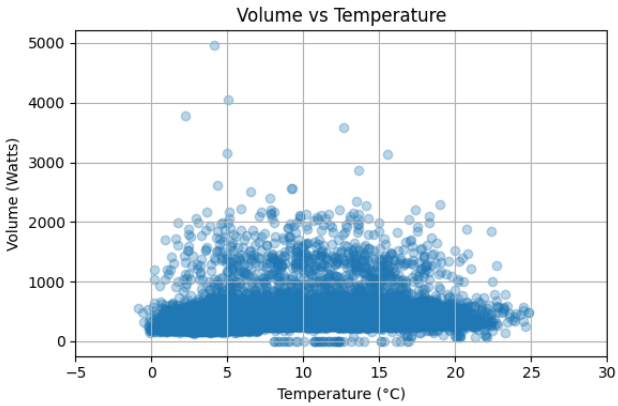
In this experiment, we analyze the different performance of our models when it comes to predicting the hourly consumption, first at a single house level and then for the whole group of households (simulating a neighborhood).

4.3.1 REFIT

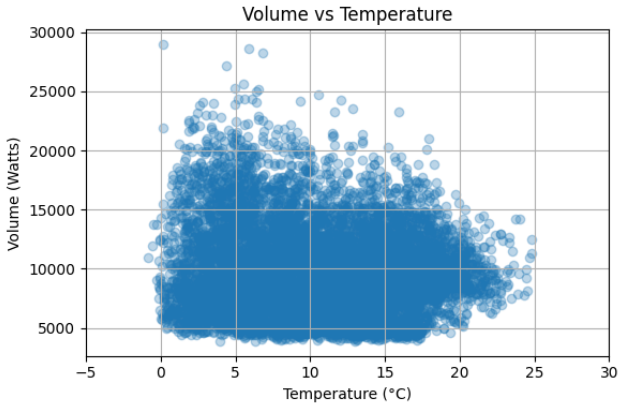
The REFIT dataset is made up of smart homes; therefore, the automated systems such as heating create much more regular and consistent patterns than if they were manually controlled. This explains why, on average, other than the shear size of the collection, we see better performance across the board than with the powerchainger set of homes.
Comparing the prediction accuracy of our models when it comes to working with one household versus many, enables us to see first hand how aggregates of homes smooth over irregular activity and therefore why it is the standard for power utilities when it comes to load forecasting.
This phenomenon can already be observed in the clustering section, as we can clearly see in table 4.2, running DBSCAN on the combined households produces a tiny number of outliers and only one cluster. As opposed to, when running DBSCAN on the homes individually, 4.1, where at minimum the algorithm found at least two clusters per house as well as a significantly larger amount of outlier days.

In order to further reinforce the argument, we can visually analyze the behavioral patterns of the aggregated households and those of our "best" individual performer. We perform this analysis from various points of view, for example, by inspecting the correlation between seasonal temperatures and load consumption, the regularity and persistence.

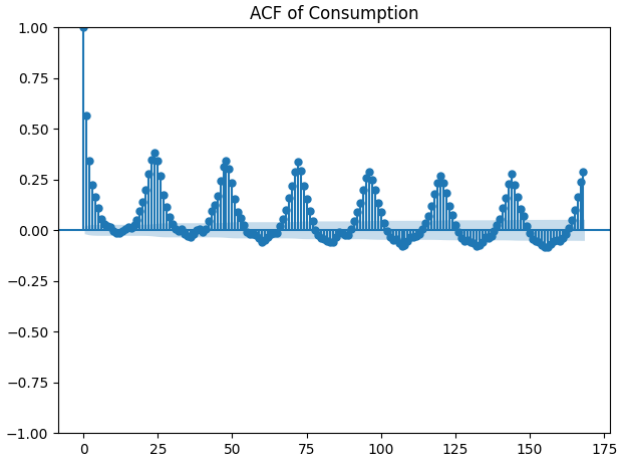
The autocorrelation function (ACF) describes how a time series is related to its own past. An ACF plot visualizes this information across a range of lags, where a lag is a on hour step



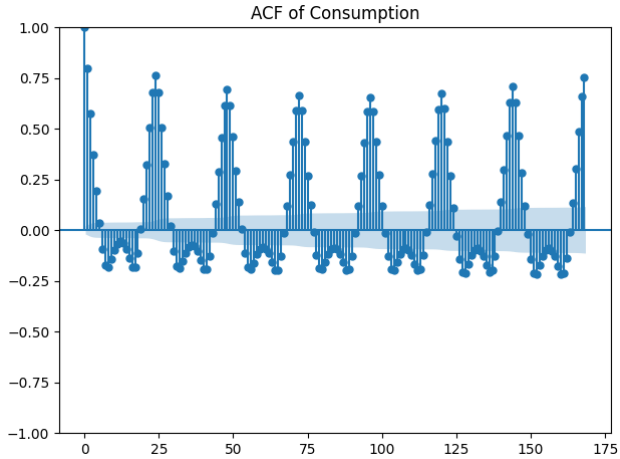
(a) House_18 : Temp vs. Consumption



(b) Aggregate : Temp vs. Consumption



(c) ACF plot for House_18



(d) ACF plot for Aggregated Houses

Figure 4.3: Behavioral patterns for House_18 vs. Aggregated households

in time. Figure 4.3, presents a sample of the visual analysis performed for house_18 and the aggregate of all the houses; namely ACF plots as well as scatter plots combining the temperature and the consumed load. When comparing our "best" representative house with the aggregated households, we can clearly see that the correlation metrics for the ACF plot are definitively more significant and more pronounced. Aggregation has strengthened autocorrelation and smoothened out irregular behavior, while much noisier, the individual household series remains seasonal. From the ACF plot for the aggregated houses, we can discern tall daily peaks, showing a string and consistent daily routine, a slow decay over time as well as clear negative troughs, indicating stable intraday variation (day and night). In contrast, the plot for house_18 still shows daily peaks, yet much less pronounced ones, indicating the presence of routines and habits but with much less consistency. We can also discern that the autocorrelation decays quicker and negative correlations are weaker, less regular intraday structure. Aggregation turns many irregular human decisions into one regular machine-like signal. When comparing the scatter plots for temperature and consumption, we note a similar trend. The aggregate shows a thick, continuous cloud, with a clear downward tendency as temperature rises, which we expect as a consequence of heating demand. This descending trend is much less discernible

on the house_18 plot, it is still visible, however, very weak and we can observe many spikes unrelated to temperature. Owing to the fact that appliances and human actions introduce noise and the possibility that heating for this household may not be entirely electric or consistently used. The following table, 4.8, shows the accuracy metrics for our models averaged out over 2 weeks of predictions for each representative house of a cluster as well as for the aggregate of all homes in the dataset.

Table 4.8: Performance metrics: Aggregate vs. Single Households

House	**ARIMA**		**XGBoost**		**LightGBM**		**Random Forest**		**Combined Mean**	
	MAE (Wh)	MAPE	MAE (Wh)	MAPE	MAE (Wh)	MAPE	MAE (Wh)	MAPE	MAE (Wh)	MAPE
house_12	204.15	53.30%	195.56	38.88%	**178.55**	**38.23%**	181.30	39.52%	183.13	38.41%
house_13	305.79	51.60%	238.65	34.84%	244.42	35.73%	**227.43**	**34.19%**	230.91	33.47%
house_18	132.74	29.13%	**106.58**	**19.96%**	111.40	20.83%	122.90	22.11%	111.99	20.65%
All Houses	2449.00	27.37%	1501.55	14.78%	**1296.88**	**13.52%**	1309.39	12.96%	1344.51	13.57%

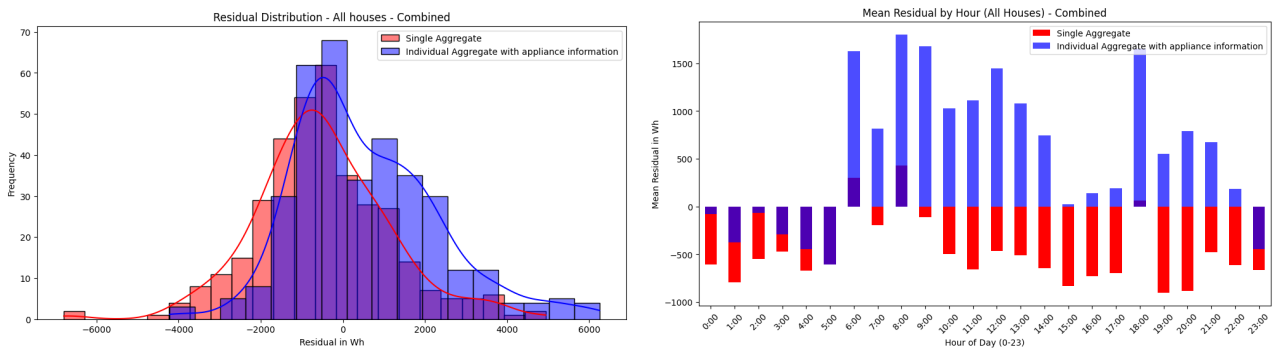
At the single household level, errors vary considerable between various users. However, the performance of our models lines up with what we would expect to see, they are the least accurate for house_12, which is part of our most irregular households. Across all households, and especially at the neighborhood level all of the ensemble models outperform the baseline, ARIMA. House_13, on the other hand, represents our slightly more regular users. In addition, this cluster of houses are our high-usage group, therefore the MAE in Watts is accordingly higher even with a slightly more accurate prediction in terms of MAPE, going from around 38-39% to 34-35%. We can clearly observe that house_18, representing our most regular and consistent households, makes for better predictions on every metric, in comparison to the previous two households, with a relative error around 20%. No single model dominates all households with LightGBM achieving the best accuracy for house_12, Random Forest for house_13 and XGBoost being the best performer when it comes to house_18. While the combined mean ensemble is competitive is does not consistently beat the best individual model, suggesting that all three algorithms make relatively similar errors. By contrast, the most accurate predictions were made for the aggregated combination of all twenty homes. The forecasting is consistently more accurate across all our models, and while the MAE might seem high, this is easily explained by the fact that the load consumed and therefore also forecasted represents that of 20 homes, consequently the MAE is expected to be at least one order of magnitude larger. This illustrates the smoothing effect of aggregation, which yields a more stable daily pattern and highlights why power utilities often focus on forecasting at the substation level.
All of these results align with our expectations, the more intriguing question is whether this holds up if we were to perform predictions for every single household and then aggregate them. As opposed to aggregating the load consumption of all houses and forecasting for the whole group at once, like was done in the previous table, 4.8. The average performance metrics for this experiment over two weeks is presented in the following table 4.9.

Table 4.9 reports the MAE in watts and the MAPE, for each ensemble model as well as the baseline statistical one, as it pertains to three different modeling strategies. All_houses, which represents a single model trained on the total neighborhood load, achieved by summing the demand of all households. The neighborhood is therefore treated by the algorithm as one very large household. Aggregated, refers to having a separate model trained for each individ-

Table 4.9: Performance metrics: Aggregate prediction vs. Aggregated predictions

Data	**ARIMA**		**XGBoost**		**LightGBM**		**Random Forest**		**Combined Mean**	
	MAE (Wh)	MAPE	MAE (Wh)	MAPE	MAE (Wh)	MAPE	MAE (Wh)	MAPE	MAE (Wh)	MAPE
All Houses	2449.00	27.37%	1501.55	**14.78%**	**1296.68**	**13.52%**	**1309.39**	**12.96%**	**1344.51**	**13.57%**
Aggregated	2757.15	27.81%	1972.47	17.18%	1563.37	15.00%	1896.86	17.02%	1781.66	16.22%
Aggregated (with Device Info)	X	X	**1418.91**	15.66%	1426.94	16.12%	1352.23	14.70%	1376.98	15.22%

ual house, using the same feature set as before, without any appliance level information. The neighborhood level forecast is constructed by summing all the household predictions. Finally, Aggregated (with Device info), as in the previous case trains a separate model for each home, however each model receives appliance features. We note that at this scale ARIMA is substantially outclassed by the machine learning models. On the first row, serving as our baseline strategy, we observe that across the three algorithms the MAE lies roughly between 1.29 kWh and 1.5 kWh, with the MAPE around 12-14%. When the modeling task is broken down to individual homes but without device information, performance is systematically degraded. The combined mean MAE rises from 1344 Wh to 1781 Wh, indicating that in the absence of extra features, training separate models introduces additional variability, likely because individual errors no longer cancel each other out as effectively. In contrast, when making use of appliance features, the models partially recover this loss. We observe clear improvements, across the board, relative to the device-free aggregated case, with the combined mean MAE dropping from around 1781 Wh to 1376 Wh. We can observe significant changes to the MAE while the relative error gains are more modest, indicating that the largest improvements occur on high-load periods. However, the models do not consistently outperform the simple total aggregate baseline. Only XGBoost shows an improvement for the absolute error (1501 Wh to 1418 Wh), however the MAPE remains slightly worse (14.78% to 15.66%), as is the case for all the metrics across every other model. Overall, these results indicate that simply decomposing the neighborhood into separate household models does not improve aggregate predictions, rather the opposite. In addition table 4.9 suggests that if no behavioral segmentation is performed, a single model on the neighborhood aggregate is hard to outperform. While augmenting individual models with device information does provide a competitive alternative, this is not a consistent observation across all models.



(a) Residual distribution in Wh

(b) Mean Residual by hour in Wh

Figure 4.4: Residual graphs for Single Model vs. Individual Models with device information

Figure 4.4, compares the distribution of forecasting residuals as well as the mean residual per hour of the day, across two modeling strategies. Namely, the single model trained on the whole neighborhood approach, represented here in red. In addition to the per household models with appliance features included, whose predictions are then summed. A forecasting residual, is defined as the actual load minus the predicted load for each hour. Therefore, if the residual is positive it indicates that the algorithm has under predicted the consumption for that hour and vice versa. Analyzing figure 4.4a, we can observe that the residuals of the neighborhood level forecasts are negatively biased and exhibit a long left tail, suggesting a systematic over prediction with occasional large errors. In contrast, the appliance informed model produces residuals that are slightly more centered around zero yet decidedly right shifted, with a large frequency between 500 and 2500 Wh. This indicates the per household approach with device information corrects the tendency to over predict, however it introduces a under prediction bias. Figure 4.4b comparing the mean residuals at every hour of the day, reveals the extent of this switch of bias. The single model approach over predicts at almost every hour, especially at night and during peak hours. On the other hand, the device informed model reduces the bias overnight, but systematically under predicts for most of the day, especially in the period from the morning to early afternoon as well as in the evening.

We can see what this same experiment looks like if we break down the houses into the cluster formed earlier in section 4.2.1, in the following table.

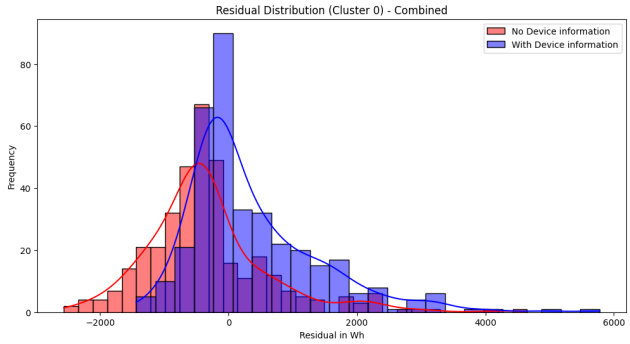
Table 4.10: Performance metrics: Aggregate prediction vs. Aggregated predictions

Cluster	**Model**	**Single Aggregate**		**Agg. (No Device Info)**		**Agg. (With Device Info)**	
		MAE	MAPE	MAE	MAPE	MAE	MAPE
Cluster 0	ARIMA	**1054.32**	45.61%	1146.00	**45.02**%	X	X
	XGB	817.63	32.63%	839.47	**30.85**%	**729.23**	36.68%
	LGBM	758.66	31.15%	768.67	**30.48**%	**754.78**	38.43%
	RF	734.58	**30.15**%	816.69	30.90%	**724.06**	33.57%
	Combined	761.62	31.18%	801.61	**30.60**%	**727.12**	35.59%
Cluster 1	ARIMA	**1229.84**	**27.92**%	1495.89	29.40%	X	X
	XGB	1037.76	20.26%	1412.91	23.85%	**947.07**	**20.00**%
	LGBM	921.97	**18.54**%	1107.09	20.99%	**853.39**	19.50%
	RF	963.27	**19.40**%	1357.26	23.44%	**874.78**	19.41%
	Combined	962.87	**19.19**%	1280.81	22.71%	**871.05**	**19.19**%
Cluster 2	ARIMA	1050.64	43.29%	**1044.57**	**41.07**%	X	X
	XGB	**391.03**	**14.24**%	404.56	14.12%	402.16	15.63%
	LGBM	**386.28**	**14.00**%	389.36	13.86%	395.45	15.75%
	RF	**391.16**	**14.23**%	423.68	14.63%	393.98	15.32%
	Combined	**386.77**	**14.05**%	398.70	13.96%	392.50	15.32%

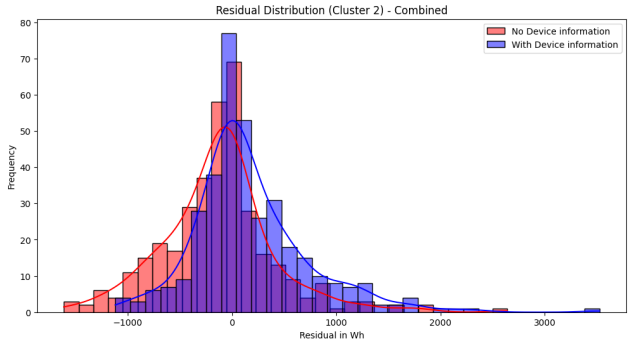
Table 4.10 compares the same three forecasting strategies that were previously explored, however this time for each user cluster. Here, Single Aggregate, refers to the first method of initially aggregating the load of each user in the cluster and then training a single model on this aggregate. Aggregate with no device information, represents the second approach of training a model for each user in the cluster then summing all the predictions. Finally, aggregate with

device information, again makes use of individual household models but with appliance features included.
For cluster 0, the most irregular users, per-household models with appliance level features systematically improves forecasting accuracy in terms of absolute error, dropping by about 5% on average compared to the total baseline. While MAPE on the other hand, slightly increases from about 31% to approximately 35%. However, as seen when evaluating the performance over all houses, the individual model approach without device information is the least effective approach. Cluster 1, shows a similar pattern, as training household models without device level information degrades the performance relative to the single aggregate model. However, once device features are included, the absolute error markedly diminishes, by around 10% om average across all three algorithms. The relative error, on the other hand, slightly improves or remains the same. Suggesting that for behaviorally complex but high use users, appliance level information delivers a significant improvement overall. By contrast, for the most regular group of users (Cluster 2), all three strategies achieve relatively low errors and the differences are small. However, adding appliance features does not provide a consistent benefit, on the contrary for all models we observe a slight worsening in both relative and absolute errors, with the combined mean MAE going from 386 Wh to 392 Wh and the MAPE from 14% to 15%. Suggesting that for highly routinized households with simple load shapes, a single aggregate model captures most of the predictable structure and that additional appliance features per-household bring little to no benefit. We also observe that for the baseline model, ARIMA, that while it is less accurate than the other models for all clusters, it shows the biggest gap for cluster 2, and the most competitive predictions for cluster 1. Taken together, table 4.10 shows a clear interaction between user type and modeling strategy, where for irregular and high-usage multi routine households (cluster 0 and 1) forecasting accuracy is improved with the addition of device information. While, for regular users (cluster 2) the simple single aggregate approach already achieves low errors and the benefits of appliance level information are marginal at best. Indicating that the choice of modeling strategy and the value of device information are dependent on the behavior of the households at hand.

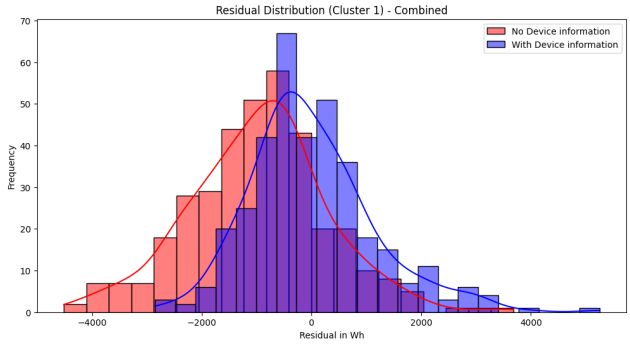
Figure 4.5, compares the distributions of hourly forecast residuals, for the three user groups, for two modeling approaches, similar to figure 4.4. It also presents the mean residual by hour of the day for Cluster 1, the most affected group. For Cluster 0, consisting of the most irregular houses, the single aggregate model produces residuals that are decidedly left shifted, demonstrating a systematic over prediction. The appliance informed approach shifts the bulk of the distribution closer to zero, reducing this bias. However, this comes at the cost of a bias towards under prediction, as indicated by the larger positive tail generated. For Cluster 2, which houses our most regular and consistent households, both strategies yield residuals that are mostly centered on zero. Similarly to Cluster 0, however less pronounced, the single aggregate model shows a slight negative bias, while the per household approach introduces a small positive shift and a more significant right tail. Confirming the marginal benefits offered by the appliance usage information for these users. Finally, Cluster 1 where the effect is most pronounced, and which consists of high usage and behaviorally complex households. The residuals generated by the single model approach are mostly negative and concentrated around large values (roughly -500 to -2000 Wh), demonstrating a strong and systematic over prediction of demand. However, the appliance informed strategy yields residuals that are centered much closer to zero, suggesting that most of the bias of the single aggregate model has been eliminated. Yet, a heavier right



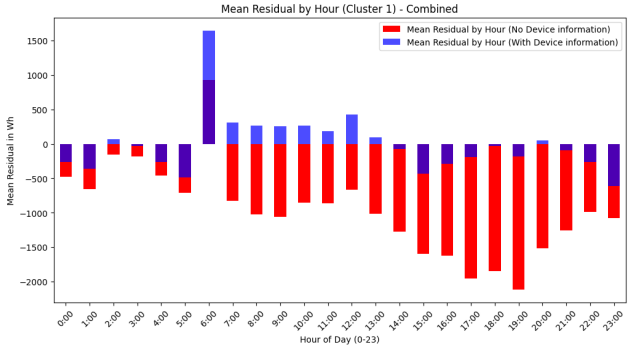
(a) Cluster 0 : Residual distribution in Wh



(b) Cluster 2 : Residual distribution in Wh



(c) Cluster 1 : Residual distribution in Wh



(d) Cluster 1 : Mean Residual by hour in Wh

Figure 4.5: Residual graphs for Single Model vs. Individual Models with device information, per cluster

side and longer positive tail remains, indicating a slight tendency towards under prediction with occasional large postive residuals. Figure 4.5d disaggregates the effects for Cluster 1 by hour of the day. The single aggregate model exhibits a negative mean residual for every single hour (except 6:00), especially in the afternoon and evening peaks, confirming the systematic over prediction of demand. The appliance informed model by contrast, attenuates this pattern with mean residuals that are much closer to zero in the evening, overnight and in the early morning. While becoming positive from morning to early afternoon. Taken together, these figures show that appliance information usefulness, is highly dependent on the type of user, showing very limited benefits for very regular households, modest gains for irregular homes and substantial improvements when correcting strong systematic bias for high usage and behaviorally complex users.

4.4 Appliance Based Predictions

In the following section, we will be presenting the experiment at the core of this project. Namely, comparing the accuracy of the predictions produced by our models with and without appliance-level information. We will focus on our selected households from the previous clustering step, and analyzing in more detail our optimal performer.

4.4.1 REFIT

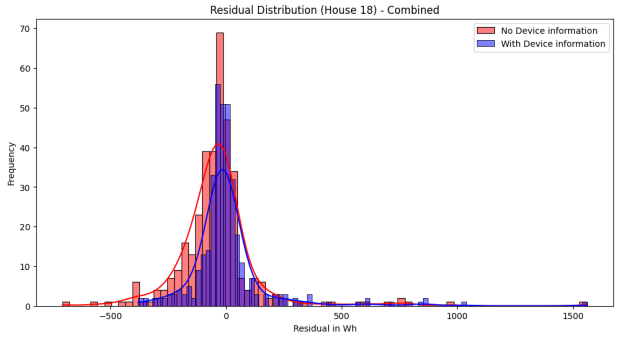
The REFIT dataset provides us with the consumption information of nine appliances for each house. However, the nine devices in questions are mostly different for every home. In this experiment, we will first be measuring the accuracy of our models while only providing them with total consumption as input. This will then be contrasted with the performance metrics of the same predictions, yet this time the models will also be provided with the device information. Table 4.11 presents a collection of performance scores averaged over fourteen days for each of the representative households.

Table 4.11: Performance comparison with and without Appliance information for single-household demand

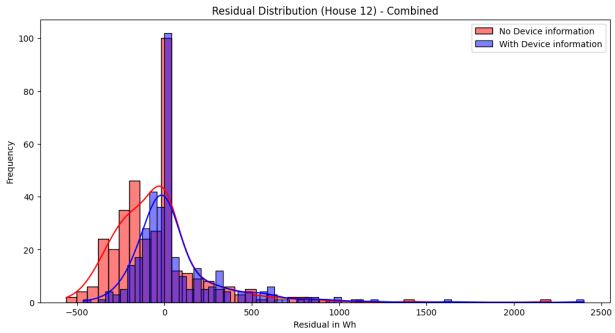
Model	**Setting**	**house_12**		**house_13**		**house_18**	
		MAE (Wh)	MAPE	MAE (Wh)	MAPE	MAE (Wh)	MAPE
ARIMA	Total Only	204.15	53.30%	305.79	51.60%	132.74	29.13%
XGBoost	Total Only	195.56	**38.88**%	238.65	**34.84**%	106.58	19.96%
	Appliance-level	**155.89**	46.45%	**206.57**	40.99%	**88.76**	**18.80**%
LightGBM	Total Only	178.55	**38.23%**	244.42	35.73%	111.40	20.83%
	Appliance-level	**156.17**	49.28%	**209.40**	**42.94**%	**88.80**	**18.67**%
Random Forest	Total Only	181.30	**39.52**%	227.43	**34.19**%	122.90	22.11%
	Appliance-level	**159.52**	45.80%	**214.41**	37.99%	**93.78**	**18.35**%
Combined Mean	Total Only	183.13	**38.41**%	230.91	**33.47%**	111.99	20.65%
	Appliance-level	**155.53**	46.23%	**208.26**	39.82%	**89.00**	**18.25**%

We can observe that all models improve with the addition of appliance information, especially when it comes to absolute error or MAE. For house_12 and house_13, the relative error consistently increases at the cost of the reduction in MAE. Although the appliance aware models reduce average absolute mistakes, they tend to perform less well in relative terms, suggesting the errors likely occur more often in low load periods as small absolute errors would translate to larger MAPEs, and in turn high-load periods likely see the most benefit. By contrast, house_18, representing regular users benefits from appliance features on both fronts, achieving lower MAE and MAPE. This indicates that for highly routinized households device information can provide consistent gains to forecasting accuracy. While the cluster level results discussed earlier in table 4.9, showed little to no effect for cluster 2, the households comprising this group are not identical. Some houses provide strongly structured appliance information such as house_18 and benefit, while others may have appliances that contribute less distinct information and therefore exhibit negative effects by contrast. The results indicate that device features help capture idiosyncratic peaks in demand, and therefore vary by user. However, the MAE gains are frequently accompanied by small increases in MAPE, suggesting that the additional information improves predictions for large loads at the cost of noisier relative error for smaller loads.

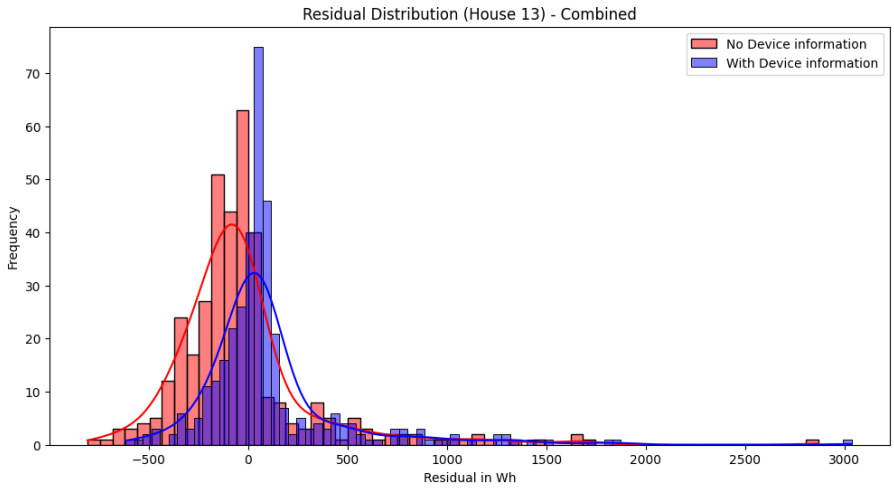
Figure 4.6, depicts the distribution of hourly forecast residuals for house_18, house_13 and house_12, belonging to Cluster 2 , 1 and 0 respectively. With respect to the two model specifications, the baseline approach without appliance level information, in red, and the augmented



(a) House 18



(b) House 12



(c) House 13

Figure 4.6: Residual distributions (Wh) comparing models without and with device information.

version with appliance features, in blue. In the case of house_18, a representative of Cluster 2, both approaches yield residuals that are relatively centered around zero and approximately symmetric. The inclusion of the appliance level information, slightly tightens the peak around zero and while attenuating the left tail, and very marginally extends the right one, indicating fewer large errors. However, the similarity of the two distributions implies that device consumption information provides limited yet clear additional improvements for this user, consistent with findings from table 4.11. As for house_12, the baseline without device information included, exhibits a distinct left shifted distribution, with the bulk of residuals residing between roughly -600 and 0 Wh. Indicating a systematic tendency to over predict, in line with the findings for the aggregate of this Cluster, with the notable presence of a few large positive residuals. By contrast, once appliance level information is made available to the model, the residual distribution moves markedly closer to zero, signaling a substantial reduction in over estimations. The appliance aware model, trades this decrease in negative residuals for a slight increase in large positive errors. For house_13, figure 4.6c, the contrast between the two approaches is much more pronounced. The baseline, in red, exhibits a strong and clear left skewed distribution, with a substantial amount of over predictions, suggesting a systematic overestimation of

demand. In contrast, when incorporating device features, the distribution tightens around a single peak close to zero. The overall distribution becomes more symmetric and concentrated, reflecting a reduction in bias as well as variance. However, just as with house_12, the approach making use of device information exhibits a slightly larger and more pronounced right tail, as it replaces a tendency to over predict with a bias towards under prediction. This illustrates that for high usage, behaviorally complex households, such as house_13 and the rest of Cluster 1, device level information can provide significant gains to the forecast accuracy of tree based models for typical levels of demand. Though these improvements are at the cost of an increase in substantial under prediction, this trade off matches the results from the previous table 4.11, where MAE is reduced while MAPE deteriorates slightly.

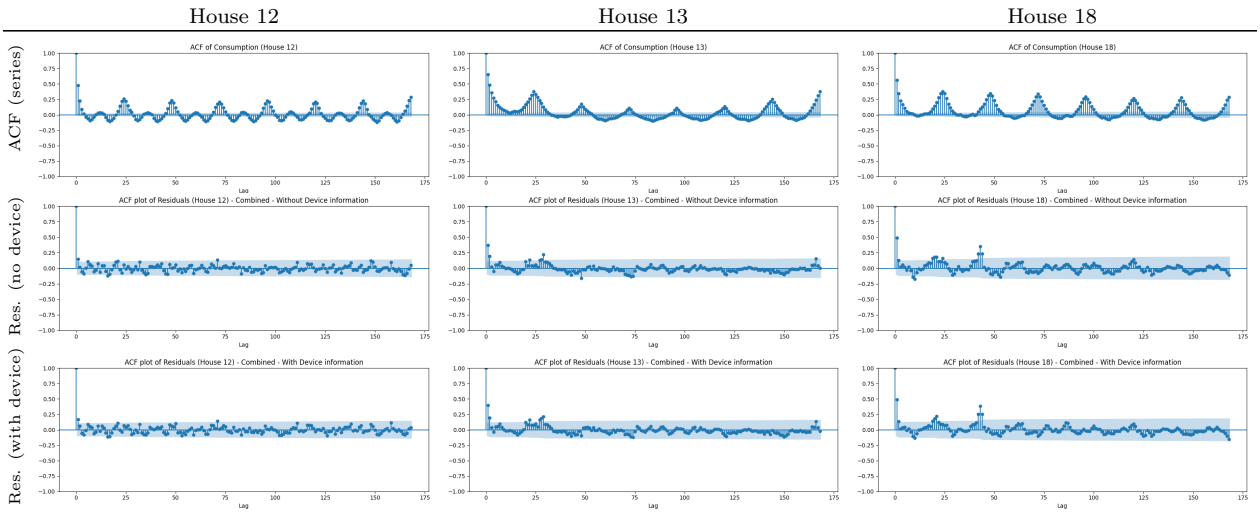


Figure 4.7: ACF of the original series (top row) and residual ACF for models without device information (middle row) and with device information (bottom row), for Houses 12, 13, and 18.

Figure 4.7, presents the Autocorrelation plots for each of the three representative houses, where each column of sub figures represents a household. Every row of sub figures corresponds initially to the ACF plot of the original time series of the household, the second row represents the the ACF plots of the residuals produced by the simple load based approach. Finally, the third row of sub figures presents the ACF plots of the residuals obtained from the appliance informed model. Across the three houses, the original load series exhibits short term dependence, daily and weekly seasonality, with the most regular patterns for house_18, in line with our expectations. For house_12, the baseline approach yields an ACF plot that is very close to white noise, with almost all correlations falling within the confidence band, suggesting that the model has captured most of the temporal structure present in the series. Once device level information is added, the residual ACF remains essentially flat and mostly unchanged. For house_13 on the other hand, some residual autocorrelation persists, especially at short lags when it comes to the consumption only model. However, once making use of appliance features, the magnitude of these short lag autocorrelations are reduced, with smaller spikes and fewer of them exceeding the confidence band. This indicates, that additional device information does indeed help explain part of the short term dependence previously left in the residuals. Finally, in the case of

house_18, the residuals produced by the baseline model are clearly not white noise, with some sharp noticeable spikes at short lags. Suggesting that despite this household's regularity, there is some remaining daily and sub-daily structure that the model fails to capture. Although some structure remains, once appliance information is integrated, most of the autocorrelations are attenuated and the plot is slightly closer to white noise. This is consistent with our findings for this house, namely the quantitative improvements to the relative and absolute error.

In the next experiment, we will dive deeper into which appliances had the most pronounced impact and how they affected the accuracy of our models.

4.4.2 Powerchainger

The dataset provided by Powerchainger includes different devices per household, some with slightly more than others. In this experiment, we will start by measuring the forecasting accuracy of our models, while not incorporating any device information. We will then be comparing these results to the performance metrics achieved using appliance information. This experiment will be making use of the two selected households in the previous clustering section, 4.2.2. However, as opposed to when working with the REFIT dataset, in section 4.4.1, the performance scores are attained over 7 days, due to the relatively small size of the Powerchainger data. Presented in table 4.12, is a summary of the accuracy metrics realized by each of our ensemble models as well as the baseline ARIMA model. Where Total-Only, refers to the strategy employing the overall consumption, while Appliance-level corresponds to the approach enhanced by appliance features.

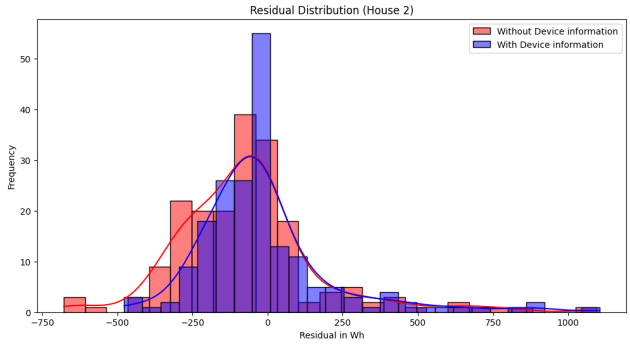
Table 4.12: Performance comparison with and without Appliance Information

Model	**Approach**	**house_2**		**house_6**		**house_9**	
		MAE (Wh)	MAPE	MAE (Wh)	MAPE	MAE (Wh)	MAPE
ARIMA	Total Only	218.93	71.95%	81.10	46.95%	114.87	36.33%
XGBoost	Total-only	186.26	55.66%	132.92	47.46%	123.49	32.72%
	Appliance-level	**149.93**	**55.44%**	**68.02**	**42.54%**	**107.46**	**31.29%**
LightGBM	Total-only	176.65	**57.28%**	106.29	45.97%	**98.90**	32.09%
	Appliance-level	**161.32**	58.63%	**71.68**	**44.12%**	99.13	**31.97%**
Random Forest	Total-only	175.46	**54.51%**	120.33	46.90%	98.68	**30.20%**
	Appliance-level	**151.04**	57.91%	**57.80**	**41.65%**	**97.90**	31.02%
Combined Mean	Total-only	178.07	**55.26%**	117.67	46.38%	101.88	**30.81%**
	Appliance-level	**153.04**	56.70%	**64.90**	**42.19%**	**99.80**	30.89%

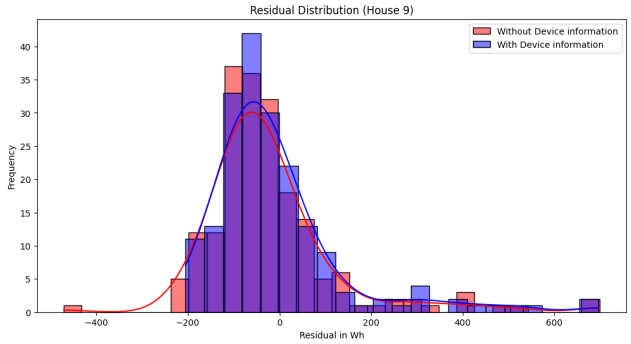
As expected, the more regular user, house_9, is easier to predict for over both strategies, which is reflected in the metrics presented in table 4.12, except for the case of ARIMA which is more accurate for house_6. Working with this smaller dataset has revealed mostly similar trends to those observed in the previous section 4.4.1. One of these trends is the fact that on these smaller datasets ARIMA performs much more competitively, especially in the case of house_9 and house_6 where it manages to outperform all of the ensemble models in terms of MAE. Due

to the small sample size of the data in question, the relative error of the models across the board is higher than for the REFIT households. However, we do observe a clear and consistent pattern in terms of absolute accuracy, for all three households every model benefits from the addition of appliance level information with the exception LightGBM for house_9 which shows a marginal negative impact. For house_2, we observe that even though MAE improves across all models, the relative accuracy slightly deteriorates, except in the case of XGBoost. This suggest that while device features do help reduce absolute deviations, they do not consistently improve proportional accuracy, likely due to periods of low consumption. By contrast, house_6 shows consistent improvements for every metric for each of the models, with significant reductions to the absolute error halving in some cases and for the combined mean dropping from around 118 Wh to 65 Wh. This indicates that for this specific user device activity is strongly aligned with demand variability and is therefore very valuable. Highlighting that while this household is not particularly regular in its usage, the behavior it exhibits can be dominated by a small number of appliances and event changes. Finally, similarly to the findings of the previous experiments on the REFIT dataset, the more regular users see less benefits appliance features. This can be discerned by house_9 which sees very modest improvements in terms of MAE with combined mean dropping from around 102 Wh to about 100 Wh, while the relative error is mostly unaffected. Suggesting that when daily patterns are relatively stable device, appliance features can help refine predictions without destabilizing them. Taken together these results support the broader conclusion that the value of device information is dependent on the behavior of the user.

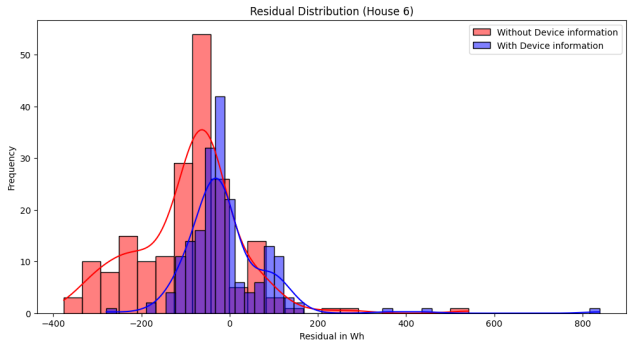
Figure 4.8, compares the distribution of the residuals produced by the combined mean model for all three houses, as well as the mean residual by hour for house_6. As in previous sections, red denotes the simple consumption only approach, while blue represents the strategy augmented by appliance features. When observing the plot for house 2, we note that it exhibits a broad and skewed distribution for both feature sets, highlighting the fact that it is the most difficult house to forecast. However, the device informed model produces a more concentrated and symmetric central mass, yet the distribution retains a pronounced right tail. These results coincide with the findings table 4.12, where the improvements in MAE may be offset by relative errors during low load periods and occasional large under predictions. On the other hand, when it comes to house 9, both approaches produce similar results, where the distributions overlap closely and show a narrow peak near zero. Adding the appliance features yields only marginal changes in the shape of the distribution as well as in the bias and spread, with the notable exception of a lighter left tail. Highlighting the small gains observed in table 4.12 for this household. By contrast, house_6 shows the clearest benefit from device information. The baseline distribution is visibly wider and more skewed towards the left with substantial frequencies away from zero. When appliance features are incorporated, the residuals are markedly more concentrated near zero, with the left tail shrinking considerably at the expense of the right tail growing slightly. This is consistent with improvements observed in both MAE and MAPE for this user, suggesting that the total load is strongly driven by identifiable appliance activity patterns that the device features allow the models to exploit. This is further reinforced by the mean residual by hour plot, where we can observe the systematic tendency to over predict from the baseline approach, with the most significant errors occurring in the late afternoon and evening. When appliance level features are introduced, the hourly residuals move closer to zero for most hours, reducing the magnitude and variability throughout the day. However, overnight the magnitude of error



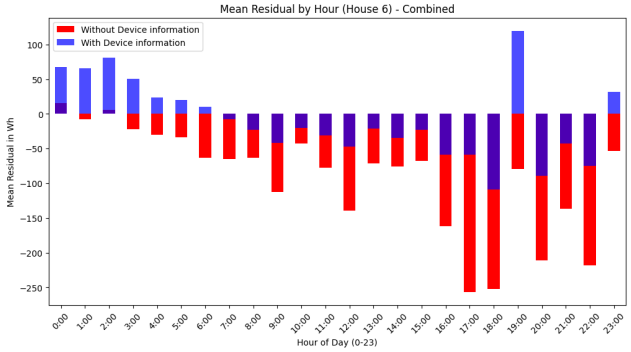
(a) House 2 : Residual Distribution



(b) House 9 : Residual Distribution



(c) House 6 : Residual Distribution



(d) House 6 : Mean Residual by hour of the day

Figure 4.8: Residual Distribution plots, per model, for House 2, 6 and 9 and mean residual by hour for House 6

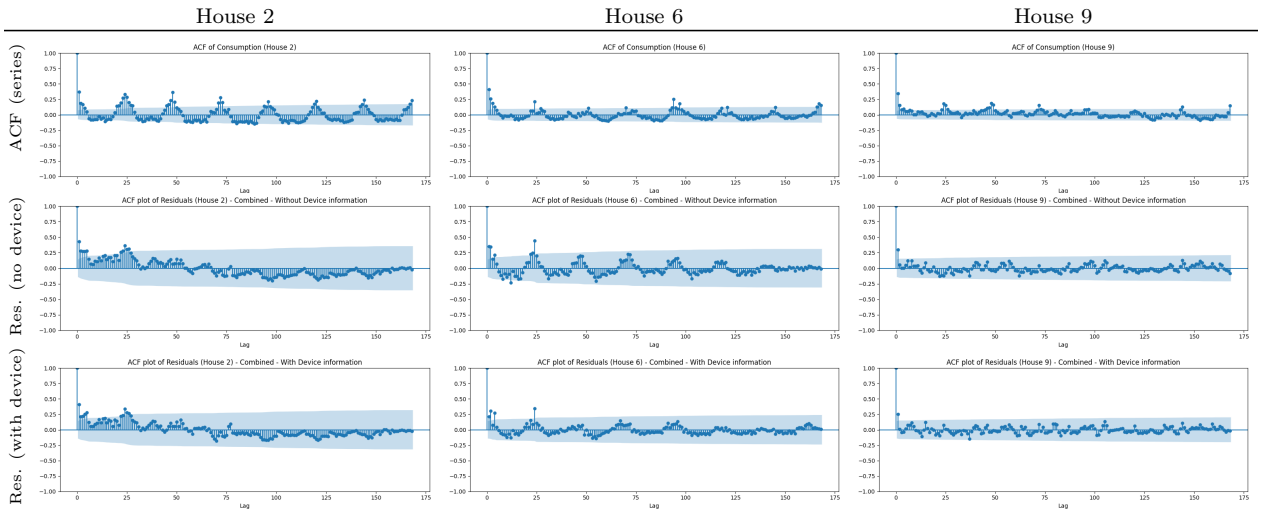


Figure 4.9: ACF of the original series (top row) and residual ACF for models without device information (middle row) and with device information (bottom row), for Houses 2, 6, and 9.

is more pronounced for the device informed strategy, suggesting that most of the gains happen during the day where total consumption is higher and explains the large improvements in absolute errors relative to the slight improvements in MAPE.

Figure 4.9, presents the ACF plots for house_2, house_6 and house_9, first for the original consumption time series, then for the residuals produced without appliance features and finally for the residuals yielded through the device enhanced approach. We can observe initially that for house_2, the original series exhibits a slowly decaying pattern with clear peaks at regular lags, representing daily cycles. House_6 shows strong short lag autocorrelation which decays rapidly, as well as visible periodic components at roughly daily intervals. However, these oscillations are smoother and less pronounced than in the case of house_2. While house_9, shows a similar yet significantly weaker structure and appears the least structured of the three. Where we can observe that for small lags the autocorrelation is clearly positive, and at larger lags the oscillations are smaller, suggesting milder daily repetition. Moving on to the ACF plot for the yielded residuals of house_2. We can discern that for the simple consumption based approach, the residuals still exhibit some positive autocorrelation at short lags, which decay gradually. This indicates that while most of the temporal pattern in the original series has been removed, some still remains. By contrast, the device features based strategy produces residuals that display slightly less autocorrelation at small lags, as well as the residuals as a whole being more tightly packed around the x-axis. In the case of house_6, the baseline strategy does not adequately capture the daily cycle, leaving a visibly periodic and highly auto-correlated set of residuals. However, once device features are incorporated the large seasonal peaks are markedly reduced and the residual ACF moves much closer to zero. This corresponds with the earlier findings where house_6 benefited substantially from appliance information. For house_9, on the other hand, both approaches yield very similar ACF plots with little to distinguish them, where aside from lag 1, the residuals are contained within the confidence bands and show no clear pattern. Adding appliance features therefore has almost no distinguishable effect, with both graphs resembling white noise, reflecting the negligible differences in MAE and MAPE discussed earlier in table 4.12. These diagnostics suggest that the value of device information is household dependent and is greatest when the forecasting errors are driven by unmodeled routine dynamics.

4.5 Device Feature Ablation

In this section we will be presenting the results of the device features ablation. Where we delve deeper into which appliance information was the most beneficial or detrimental to the accuracy of our models. In order to attain that goal we will be conducting a feature ablation study focused on the appliances available, where we test every possible combination of said appliances. This is done by computing every possible permutations of devices with the same set of model parameters. We chose to use XGBoost for this experiment as over the previous experiments it has benefited the most on average from the addition of appliance features.

The selected model parameters are the following:
- n_estimators = 200
- max_depth = 6
- learning_rate = 0.05
- subsample = 1.0
- colsample_bytree = 0.8

4.5.1 REFIT

Building upon the work done in the previous experiment, more precisely in section 4.4.1, we will explore the effects of the various appliances of the three representative households. We have identified the following best configurations for each of the homes.

Table 4.13: Best-performing appliance combinations for House_12

Rank	Appliance Combination	Mean MAE (Wh)
–	All Appliances (Baseline)	162.18
1	Television (Lounge), Kettle	160.68
2	Television (Lounge), Microwave, Kettle	161.95
3	Fridge-Freezer, TV (Lounge), Microwave, Kettle, Toaster, TV (Bedroom)	162.18
4	Television (Lounge), Microwave, Kettle, Toaster	162.27
5	Fridge-Freezer, TV (Lounge), Kettle, Toaster, TV (Bedroom)	162.74

Table 4.13, summarizes five top performing combinations of appliances for house_12, as well as the baseline (All appliances). We can observe that several combinations perform just as well or even slightly better than the baseline. All of the top configurations include the lounge television and kettle, while other appliances have very little effect and in some cases hinder the performance. With the best combination of appliances being solely comprised of those two devices.

Table 4.14: Marginal contribution analysis for House_12

Appliance	Marginal Mean (Wh)	Marginal Std (Wh)
Television (Lounge)	−20.87	13.52
Kettle	−18.45	12.48
Fridge-Freezer	−5.14	9.27
Microwave	−4.38	8.05
Toaster	1.09	7.79
Television (Bedroom)	1.09	8.24

Table 4.14, reports the results of the marginal contribution analysis for house_12. We observe as was hinted at earlier in table 4.13, that two appliances have the most significant impact, Television (Lounge) and Kettle, with contributions of -20.87 Wh and -18.5 Wh respectively. The marginal standard deviation for both of these devices also highlight that they consistently

have a positive impact on the accuracy of our models. Other appliances such as Fridge-Freezer and Microwave have modest marginal contributions. While, Toaster and Television (Bedroom) exhibit slightly positive marginal contributions, suggesting that they add noise rather than information for this household. The relatively large standard deviation of the appliances hint at the fact that the effect of every appliance here is context dependent. Overall, for this irregular user, a small subset of frequently used appliances provides the most benefits.

Table 4.15: Best-performing appliance combinations for House_13

Rank	**Appliance Combination**	**Mean MAE (Wh)**
–	All Appliances (Baseline)	206.57
1	Washing Machine, Dishwasher, Tumble Dryer, Computer, Kettle	195.99
2	TV (1), Washing Machine, Dishwasher, Tumble Dryer, Computer, Microwave	196.08
3	Unknown, Washing Machine, Dishwasher, Tumble Dryer, Computer, Microwave	196.34
4	Unknown, Washing Machine, Dishwasher, Tumble Dryer, Computer, Microwave, Kettle	196.49
5	TV (1), Unknown, Dishwasher, Tumble Dryer, Computer, Kettle	196.62

Table 4.15, provides an overview of the top five performing configurations of appliances. Unfortunately for house_13, one of the devices present in the dataset is labeled "Unknown". As opposed to house_12, this household exhibits a more significant reduction in MAE for the best performers when compared to the baseline, from 206.57 Wh to 196 Wh, roughly a 5% reduction. We can make out that all of the top combinations include a core set of three appliances : Tumble Dryer, Dishwasher and Computer site. In addition to a combination of Washing Machine, either Microwave or Kettle and sometimes "Unknown" or Television Site 1. Hinting that, for this household, a limited group of high-energy devices carry most of the useful additional information.

Table 4.16: Marginal contribution analysis for House_13

Appliance	**Marginal Mean (Wh)**	**Marginal Std (Wh)**
Tumble Dryer	−11.15	9.08
Computer	−3.24	5.27
Dishwasher	−2.14	5.82
Kettle	−1.50	5.47
Television (2)	−1.27	9.58
Unknown	−0.39	4.37
Washing Machine	−0.31	5.15
Microwave	−0.14	4.73
Television (1)	0.03	4.38

Summarized in table 4.16, are the marginal contributions of all the appliances available for house_13. The device with strongest average effect is the Tumble Dryer with a marginal mean of -11.5 Wh, and standard deviation of about 9 Wh, indicating that this appliance essentially always has a positive impact. Followed by Computer Site, Dishwasher and Kettle, with marginal means of -3.24, -2.14 and -1.50 respectively. While other devices have smaller

and mostly insignificant marginal contributions to the MAE, such as Television Site 2. The final appliances exhibit marginal MAEs very close to zero and therefore have a negligible average effect. Compared to the previous such table for house_13, table 4.14, the marginal standard deviations are relatively moderate, indicating that these rankings are mostly stable. All in all, we can say that for this high-usage and somewhat irregular household, appliance level information is most helpful when restricted to a subset of high-usage routinely used devices, such as those related to laundry, computing or the kitchen. Making use of all appliance features is not optimal and a careful selection will yield more accurate predictions.

Table 4.17: Best-performing appliance combinations for House_18

Rank	Appliance Combination	Mean MAE (Wh)
–	All Appliances (Baseline)	88.03
1	Washing Machine, Dishwasher, Desktop Computer, Microwave	88.50
2	Washer Dryer (Garage), Dishwasher, Desktop Computer, Microwave	88.70
3	Fridge (Garage), Washer Dryer (Garage), Washing Machine, Dishwasher, Desktop Computer, Microwave	88.73
4	Washer Dryer (Garage), Washing Machine, Dishwasher, Desktop Computer, Microwave	88.82
5	Fridge (Garage), Washing Machine, Dishwasher, Desktop Computer, Microwave	89.30

Presented in table 4.17, are the five top performing combinations of appliances ranked by mean MAE, as well as the baseline. Immediately we note that in contrast to the two other households, the baseline (including all appliances) is the best configuration in terms of average MAE, with 88.03 Wh. All reduced combinations, exhibit very slightly higher MAEs, ranging from 88.5 Wh to 89.30 Wh. Indicating that removing any subset of appliances hinders the accuracy of the predictions and at best leave the performance essentially unchanged.

Table 4.18: Marginal contribution analysis for House_18

Appliance	Marginal Mean (Wh)	Marginal Std (Wh)
Dishwasher	−8.99	5.29
Television	−3.61	5.40
Desktop Computer	−2.98	3.28
Washing Machine	−2.66	3.40
Microwave	−2.08	3.47
Washer Dryer (Garage)	−0.75	3.21
Fridge-Freezer	−0.65	3.30
Fridge (Garage)	−0.34	2.88
Freezer (Garage)	0.59	3.26

The marginal contribution results, presented in table 4.18, offer a more nuanced picture of the individual effect of each of the available appliances. The most notable device for this house is the Dishwasher with a negative marginal mean of -8.99 Wh, a marginal standard deviation of 5.29 Wh, suggesting that it almost always has a positive effect on accuracy. Multiple other appliances also offer slight improvements, from Television Site to Microwave, with marginal means ranging from -3.61 Wh to -2.08 Wh. While the rest of the devices offer either a very small decrease in MAE, a negligible impact close to zero, or even a very slight worsening of the accuracy. However, the magnitudes for each are generally modest and no single appliance

dominates to the same extent as with the two previous households. Overall, these results indicate that for the most regular users, the impact of device features is spread relatively evenly, and the models are able to make use of the full set of appliances without overfitting. Consequently, each appliance offers at least small but cumulative pieces of information. These findings line up with the previously observed trends and the fact that appliance level information has different effects based on the user in question. When it comes to irregular and high-usage households, targeted appliance selection is more beneficial, as it is important in order to avoid adding noise. While for regular users, a more complete set of devices is more effective as the models are able to make use of most of the additional information.

4.5.2 Powerchainger

Following in the footsteps of the previous section 4.5, we will present the results of the device feature ablation study conducted with the two selected households from the Powerchainger dataset. This experiment was performed in much the same way as with the REFIT houses.

Table 4.19: Best-performing appliance combinations for House_2

Rank	Appliance Combination	Mean MAE (Wh)
–	All Appliances (Baseline)	164.35
1	Water Boiler, Oven, Washing Machine, Electric Heater	154.92
1	Water Boiler, Oven, Washing Machine, Dryer	154.92
3	TV, Water Boiler, Washing Machine, Refrigerator	156.47
4	TV, Water Boiler, Washing Machine, Electric Heater	156.53
4	TV, Water Boiler, Washing Machine, Dryer	156.53

Table 4.20: Marginal contribution analysis for House_2

Appliance	Marginal Mean (Wh)	Marginal Std (Wh)
Water Boiler	−29.75	6.46
Washing Machine	−8.53	6.12
TV	−2.65	6.52
Oven	−1.10	5.60
Electric Heater	−1.06	5.16
Dryer	−1.06	5.16
Refrigerator	−0.45	5.14

Table 4.19, reports the best performing combinations of appliances for house 2. The baseline configuration, which includes all available appliances, yields a mean MAE of 164.35 W. However, all top five combinations of subsets of devices outperform this baseline quite substantially, ranging from 154.92 Wh to 156.53 Wh. All of the best listed configurations make use of Waterboiler and Washing Machine. These findings are also supported by the marginal contributions identified for each, in table 4.20, where we can notice that they are by far the two leading contributors. The marginal mean exhibited by Waterboiler is by a wide margin the

most substantial, and nearly always provides a positive effect on the accuracy of predictions. Despite offering a less pronounced impact, other appliances such as: TV, Oven, Electric Heater and Refrigerator, make appearances in the best configurations. This is in line with the findings in table 4.20, where we can observe that these devices exhibit negative marginal means, however, these contributions are modest and range from -2.65 Wh to -1.06 Wh. Finally, Refrigerator has a very small, almost negligible, effect on the performance of the models. Overall, these results indicate that most of the additional benefits provided by appliance features, are concentrated on a small set of high impact devices, namely Waterboiler and to a lesser extent, Washing Machine. Making use of all available appliances is not necessarily advantageous and can slightly worsen forecast accuracy, by introducing extra noise. These findings align with the observation made in section 4.4.2, where device level information for this household reduced the absolute error but did not significantly alter the shape of the residuals' distribution, and did very slightly increase the relative error.

Table 4.21: Best-performing appliance combinations for House_6

Rank	**Appliance Combination**	**Mean MAE (Wh)**
–	All Appliances (Baseline)	76.22
1	Kettle, Toaster, Washer/Dryer, TV	70.77
2	Kettle, Washer/Dryer, Heatpump, TV	71.04
3	Kettle, Toaster, Washer/Dryer, Coffee Machine, Heatpump	71.23
4	Kettle, Toaster, Washer/Dryer, Coffee Machine, TV	72.81
5	Kettle, Toaster, Washer/Dryer, Coffee Machine	73.23

Table 4.22: Marginal contribution analysis for House_6

Appliance	**Marginal Mean (Wh)**	**Marginal Std (Wh)**
Washer/Dryer	−50.74	7.79
Kettle	−8.83	7.45
Coffee Machine	−3.45	7.45
TV	−2.99	6.76
Toaster	−2.61	6.67
Heatpump	−2.45	7.09

Tables 4.21 and 4.22 summarize the device combination search and the marginal contribution analysis respectively, for house_6. Where we observe that several combinations outperform the baseline, with the best (Kettle, Washer/Dryer, Heatpump and TV) achieving about 6 Wh in reduction in absolute terms. Highliting that the addition of the information for some devices may have a negative impact on accuracy. Recurrent in all of the top five best performing combinations are Kettle and Washer/Dryer, suggesting that these devices capture the dominant repeatable appliance driven patterns for this household. This is further verified by the marginal contribution results in table 4.22, where Washer/Dryer stands out as the most influential input, with a very strong negative marginal mean of about -51 Wh. Although the accompanying standard deviation is of about 7.8 Wh, the magnitude of the mean effect is enough to confirm

the robustness of the contribution. The rest of the devices in this household all exhibit negative marginal means, implying that they all contribute positively to the prediction accuracy, however, the size of this effect varies depending on the specific subset and time window. Overall, the results for this house indicate that appliance level information is most beneficial when restricted to a subset of high impact, routinely used devices. Particularly those with distinct high power events, such as Washer/Dryer, and/or regular short duration usage, like Kettle or Coffee Machine. These findings are consistent with earlier result from section 4.4.2, where we observed improvements to MAE as well as to the residual autocorrelation structure, once device features were introduced which appear to have been primarily driven by a subset of appliances with strong predictive activation patterns.

Table 4.23: Best-performing appliance combinations for House_9

Rank	**Appliance Combination**	**Mean MAE (Wh)**
–	All Appliances (Baseline)	108.61
1	Kettle, TV	104.17
2	TV, Production	104.72
3	Washing Machine, TV, Production	104.85
4	Toaster, Kettle, TV, Production	105.80
5	Washing Machine, Toaster, Kettle	106.16

Table 4.24: Marginal contribution analysis for House_9

Appliance	**Marginal Mean (Wh)**	**Marginal Std (Wh)**
TV	−4.45	3.65
Washing Machine	−1.93	3.69
Kettle	−1.81	4.73
Production	0.39	4.97
Toaster	1.21	4.99

We depict the best device combinations for house 9 in table 4.23, where we identify Kettle and TV as the best configuration, moderately outperforming the baseline by roughly 4.44 Wh. A recurring device in the top four best performing combinations is TV, indicating that it provides a consistent positive effect on the forecast accuracy. This fact is further reinforced in table 4.24, which highlights that TV indeed offers the most significant impact to MAE, with a marginal mean of -4.45 Wh. While having the most pronounced effect, it's relatively high standard deviation of 3.65 Wh indicates that this impact is quite modest. Washing Machine and Kettle are the next two most important contributors to the performance of our models, however the effect of these appliances is relatively modest, with average contributions of -1.93 and -1.81 Wh respectively. By contrast, Production and Toaster exhibit small positive marginal means, indicating that on average they hinder the prediction accuracy. Overall, the relatively high standard deviations (ranging from 3.65 Wh to 4.99 Wh) compared to the mean values (ranging from -4.45 Wh to 1.21 Wh), suggests that the effects of these devices varies significantly across iterations. These results are consistent with earlier findings from section

4.4.2, where we observed very minimal overall improvements in the performance of the models for house 9. Generally, the results for these households with limited data, indicates that adding appliance features is most beneficial when they are restricted to a subset consisting mainly of high impact routinely used devices, such as TVs, Washing Machines, Kettles and Waterboilers.

Chapter 5

Conclusion & Future work

This work investigated wether appliance level information provides measurable benefits for short term residential electricity demand forecasting when it comes to ensemble models. Two main objectives motivated this study, namely analyzing the impact of device features on ensemble models, and the comparison of a couple appliance aware tree based methods.
Across the experiments, household electricity demand has been shown to be highly heterogeneous. A consistency analysis using DBSCAN, conducted by clustering daily profiles, quantified this variability through two interpretable metrics, the number of recurring day types (DBSCAN clusters) and the fraction of atypical days (DBSCAN outliers). This yielded behavior profiles ranging from regular households with few day types and few outliers, as well as irregular ones exhibiting many day types and high outlier fractions. These descriptors were only weakly related to mean daily usage, supporting the view that predictability is mainly driven by routine regularity rather that consumption level alone.
Prediction results demonstrated that the value of device level features varies and is context dependent. At the level of single household forecasting, appliance information consistently reduced absolute error, however the effect on the relative error was more variable. In particular some irregular households exhibited heavier residual tails or increased MAPE when incorporating appliance features, indicating that device information can improve typical errors while still leaving mostly low load proportional errors and some occasional large under predictions. In contrast, more regular households tended to see improvements to both the MAE and MAPE as well as reduced residual autocorrelation, suggesting that appliance activity can act as a stable explanatory signal that helps whiten model errors. At the aggregate level, two strategies were evaluated, forecasting the aggregate load directly with a single model, and summing individual household predictions. When all households were pooled without including appliance information, forecasting the aggregate load directly proved to be significantly more accurate. Once appliance features were included the models proved to be more competitive yet were on average less precise. On the other hand, when households were segmented into behavior driven clusters, the benefits of device information became clearer. For irregular and multi routine groups, adding appliance level features typically improved performance relative to aggregate only baselines and reduced residual autocorrelation. For regular clusters, baseline models already captured most temporal structure and device information contributed smaller or inconsistent gains. One clear and consistent trend observed at both the aggregate and single household level, was a bias towards over prediction without appliance information, which once

included was replaced by a tendency towards under predicting.
In terms of model comparison, XGBoost, LightGBM and Random Forest achieved and displayed broadly similar performance and trends, with no single method consistently outperforming the rest across all experiments. The simple ensemble (mean combination) was generally competitive and provided robust performance over a range of diverse conditions. All ensemble models consistently outperformed the classical statistical baseline, ARIMA, except in the case of smaller datasets where ARIMA proved to be competitive. Finally, the device ablation experiment showed that more devices is not necessarily desired, for most households a smaller subset of high impact appliances accounted for most of the predictive benefit. While additional devices contributed marginally or introduced noise. This is consistent with the observation that appliance features are most effective when they represent stable, repeated behaviors rather that sparse informative signals.
Overall, the results support three main conclusions. Firstly, behavioral segmentation is essential to understanding and fairly evaluating appliance aware forecasting, aggregated results can hide both strong gains for some users and negligible or mixed effects for others. Secondly, device information can materially improve ensemble produced predictions for specific households and behavior driven clusters, in particular when residual autocorrelation indicates a remaining structure without appliance features. Finally, ensemble models can be made device aware in a straightforward manner, however the positive effects depend on both the user's behavior regularity and the selection of appliances rather than on the underlying method.

This study would benefit from future work, such as more complex and granular behavioral clustering. Richer representations of daily profiles and behavior types could yield more robust and consistent findings. Furthermore, a principled selection of appliance features would be worth exploring in order to maximize the performance gains. Additionally, future work should consider repeating the experiment over longer windows of data points, in order to yield more robust evaluation and study the stability and persistence of device feature benefits under changing conditions. These extensions amongst others would likely produce more consistent and clear guidance for practical adoption.

Bibliography

[1] E. Commission and D.-G. for Climate Action, *Climate action progress report 2025 – Strengthening competitiveness on the road to climate neutrality.* EU publications, 2025.

[2] Y. Mou, A. Papavasiliou, K. Hartz, A. Dusolt, and C. Redl, "An analysis of shortage pricing and capacity remuneration mechanisms on the pan-european common energy market," *Energy Policy*, vol. 183, p. 113843, 2023.

[3] J. N. Tsitsiklis and Y. Xu, "Pricing of fluctuations in electricity markets," *European Journal of Operational Research*, vol. 246, no. 1, pp. 199–208, 2015.

[4] C. B. voor de Statistiek (CBS), "Electricity consumption of domestic transport nearly doubles," Jul 2025.

[5] M. Bernecker, M. Gebhardt, S. B. Amor, M. Wolter, and F. Müsgens, "Quantifying the impact of load forecasting accuracy on congestion management in distribution grids," *International Journal of Electrical Power & Energy Systems*, vol. 168, p. 110713, 2025.

[6] M. N. Fekri, K. Grolinger, and S. Mir, "Distributed load forecasting using smart meter data: Federated learning with recurrent neural networks," *International Journal of Electrical Power & Energy Systems*, vol. 137, p. 107669, 2022.

[7] F. Rodrigues, C. Cardeira, J. M. F. Calado, and R. Melicio, "Short-term load forecasting of electricity demand for the residential sector based on modelling techniques: A systematic review," *Energies*, vol. 16, p. 4098, May 2023.

[8] S. Akhtar, S. Shahzad, A. Zaheer, H. S. Ullah, H. Kilic, R. Gono, M. Jasiński, and Z. Leonowicz, "Short-term load forecasting models: A review of challenges, progress, and the road ahead," *Energies*, vol. 16, p. 4060, May 2023.

[9] T. Hong and M. Shahidehpour, "Load forecasting case study," tech. rep., Eastern Interconnection States' Planning Council (EISPC) and U.S. Department of Energy, January 2015. A primer and in-depth discussion on load forecasting for utility commissioners and researchers.

[10] B. W.-K. Yiu, T. Zhang, and C.-W. Lee, "Short-term load forecasting using regularized greedy forest-based ensemble model," *IEEE Access*, vol. 12, pp. 112426–112439, 2024.

[11] E. Chodakowska, D. Nazarko, and Łukasz Nazarko, "Arima models in electrical load forecasting and their robustness to noise," *Energies*, vol. 14, p. 7952, Nov. 2021.

[12] S. Karamolegkos and D. E. Koulouriotis, "Advancing short-term load forecasting with decomposed fourier arima: A case study on the greek energy market," *Energy*, vol. 325, p. 135854, 2025.

[13] T. Osgonbaatar, P. Matrenin, M. Safaraliev, I. Zicmane, A. Rusina, and S. Kokin, "A rank analysis and ensemble machine learning model for load forecasting in the nodes of the central mongolian power system," *Inventions*, vol. 8, no. 5, 2023.

[14] E. A. Madrid and N. Antonio, "Short-term electricity load forecasting with machine learning," *Information*, vol. 12, no. 50, 2021.

[15] S. Hadri, Y. Naitmalek, M. Najib, M. Bakhouya, Y. Fakhri, and M. Elaroussi, "A comparative study of predictive approaches for load forecasting in smart buildings," *Procedia Computer Science*, vol. 160, pp. 173–180, 2019. The 10th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN-2019) / The 9th International Conference on Current and Future Trends of Information and Communication Technologies in Healthcare (ICTH-2019) / Affiliated Workshops.

[16] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," *CoRR*, vol. abs/1603.02754, 2016.

[17] P. Zhang, Y. Jia, and Y. Shang, "Research and application of xgboost in imbalanced data," *International Journal of Distributed Sensor Networks*, vol. 18, no. 6, p. 15501329221106935, 2022.

[18] M. Niazkar, A. Menapace, B. Brentan, R. Piraei, D. Jimenez, P. Dhawan, and M. Righetti, "Applications of xgboost in water resources engineering: A systematic literature review (dec 2018–may 2023)," *Environmental Modelling & Software*, vol. 174, p. 105971, 2024.

[19] A. Ogunleye and Q.-G. Wang, "Xgboost model for chronic kidney disease diagnosis," *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol. 17, no. 6, pp. 2131–2140, 2020.

[20] M. Ma, G. Zhao, B. He, Q. Li, H. Dong, S. Wang, and Z. Wang, "Xgboost-based method for flash flood risk assessment," *Journal of Hydrology*, vol. 598, p. 126382, 2021.

[21] H. I. Beloev, S. R. Saitov, A. A. Filimonova, N. D. Chichirova, O. E. Babikov, and I. K. Iliev, "Short-term electrical load forecasting based on xgboost model," *Energies*, vol. 18, no. 5144, 2025.

[22] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "Lightgbm: A highly efficient gradient boosting decision tree," in *31st Conference on Neural Information Processing Systems (NIPS 2017*, 12 2017.

[23] S. S. Biswas, A. Kundu, and J. Chakrabartty, "Lightgbm-driven one-day-ahead electrical load forecasting for enhancing operational efficiency of a power distribution company of bangladesh," *Electrical Engineering*, vol. 107, pp. 10379 – 10402, 2025.

[24] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 10 2001.

[25] Z. Wang, Y. Wang, R. Zeng, R. S. Srinivasan, and S. Ahrentzen, "Random forest based hourly building energy prediction," *Energy and Buildings*, vol. 171, pp. 11–25, 2018.

[26] H. Rathore, H. K. Meena, and P. Jain, "Prediction of ev energy consumption using random forest and xgboost," in *2023 International Conference on Power Electronics and Energy (ICPEE)*, pp. 1–6, 2023.

[27] R. Banik and A. Biswas, "Enhanced renewable power and load forecasting using rf-xgboost stacked ensemble," *Electrical Engineering*, vol. 106, pp. 1–21, 02 2024.

[28] M. Razghandi and D. Turgut, "Residential appliance-level load forecasting with deep learning," in *GLOBECOM 2020 - 2020 IEEE Global Communications Conference*, pp. 1–6, 2020.

[29] A. Langevin, M. Cheriet, and G. Gagnon, "Efficient deep generative model for short-term household load forecasting using non-intrusive load monitoring," *Sustainable Energy, Grids and Networks*, vol. 34, p. 101006, 2023.

[30] D. Murray, J. Liao, L. Stankovic, V. Stankovic, R. Hauxwell-Baldwin, C. Wilson, M. Coleman, T. Kane, and S. Firth, *A data management platform for personalised real-time energy feedback*, ch. IET, p. GBR. Publications Office of the European Union, 2015.

[31] D. Murray, L. Stankovic, and V. Stankovic, "An electrical load measurements dataset of united kingdom households from a two-year longitudinal study," *Scientific Data*, vol. 4, p. 160122, 01 2017.

[32] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*, KDD'96, p. 226–231, AAAI Press, 1996.

[33] W. Kong, Z. Y. Dong, Y. Jia, D. J. Hill, Y. Xu, and Y. Zhang, "Short-term residential load forecasting based on lstm recurrent neural network," *IEEE Transactions on Smart Grid*, vol. 10, no. 1, pp. 841–851, 2019.

[34] L. Yin, H. Hu, K. Li, G. Zheng, Y. Qu, and H. Chen, "Improvement of dbscan algorithm based on k-dist graph for adaptive determining parameters," *Electronics*, vol. 12, no. 15, 2023.

[35] J. Friedman, "Greedy function approximation: A gradient boosting machine," *The Annals of Statistics*, vol. 29, 11 2000.